

Chemistry Practice Questions — 150 Problems

Beginner (C1–C50): Level 1–2, 1–3 cards. Intermediate (C51–C100): Level 2–3, 2–5 cards. Advanced (C101–C150): Level 3–4, 4–10 cards.

Every [Cards: ...] tag is an authoritative reference to [honors-chemistry-expression-to-equation-vocabulary.md](#). Solutions in [chemistry-answers.md](#). Method: [chemistry-without-intuition.md](#).

BEGINNER — 50 Problems (Level 1–2, 1–3 cards)

C1–C8: Stoichiometry — Mole Bridge

- C1.** How many moles are in 88.0 g of CO_2 ? ($M_{\text{CO}_2} = 44.0 \text{ g/mol}$) [Cards: 1]
- C2.** What is the mass of 2.50 mol of H_2O ? ($M_{\text{H}_2\text{O}} = 18.0 \text{ g/mol}$) [Cards: 1]
- C3.** How many molecules are in 0.500 mol of N_2 ? [Cards: 4, 5]
- C4.** What volume does 3.00 mol of O_2 occupy at STP? [Cards: 6, 7]
- C5.** Balance and give the mole ratio of H_2 to NH_3 : $\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$. [Cards: 8, 9]
- C6.** How many grams of NaCl are produced from 23.0 g of Na reacting with excess Cl_2 ? ($M_{\text{Na}} = 23.0$, $M_{\text{NaCl}} = 58.5$) [Cards: 1, 8, 10]
- C7.** How many moles of H_2O are produced when 4.00 mol of O_2 react with excess H_2 ? ($2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$) [Cards: 8, 9]
- C8.** Find the percent composition of carbon in CH_4 . ($M_{\text{C}} = 12.0$, $M_{\text{CH}_4} = 16.0$) [Cards: 17]

C9–C14: Gases

- C9.** Find the pressure exerted by 2.00 mol of an ideal gas in a 10.0 L container at 300 K. ($R = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$) [Cards: 19, 20]
- C10.** A sealed 5.00 L container of gas at 1.20 atm and 300 K is heated to 400 K. What is the new pressure? [Cards: 21]
- C11.** A balloon at 2.00 atm and 300 K has volume 3.00 L. It rises to where pressure is 1.50 atm and temperature is 280 K. New volume? [Cards: 21]
- C12.** A gas at constant temperature has its pressure doubled. What happens to its volume? [Cards: 22]
- C13.** A gas at constant pressure is cooled from 400 K to 200 K. Its initial volume is 8.00 L. Final volume? [Cards: 23]

C14. Equal volumes of H_2 and O_2 at the same T and P contain equal numbers of moles. Which card guarantees this? [Cards: 25]

C15–C20: Thermochemistry Basics

C15. How much heat is required to raise 50.0 g of water from 20.0°C to 80.0°C? ($c_w = 4.184 \text{ J g}^{-1} \text{ K}^{-1}$) [Cards: 33, 34]

C16. A reaction releases 250 kJ of heat to the surroundings. Is it exothermic or endothermic? What is the sign of ΔH ? [Cards: 39, 40, 41]

C17. A 100.0 g aluminum block ($c = 0.900 \text{ J g}^{-1} \text{ K}^{-1}$) cools from 150.0°C to 25.0°C. How much heat is released? [Cards: 33]

C18. In a coffee-cup calorimeter, 50.0 g of hot water at 60.0°C is mixed with 50.0 g of cold water at 20.0°C. Predict the final temperature. ($c_w = 4.184$) [Cards: 33, 34, 36]

C19. The specific heat of water is $4.184 \text{ J g}^{-1} \text{ K}^{-1}$. What is the molar heat capacity of water? ($M = 18.0 \text{ g/mol}$) [Cards: 34, 38]

C20. A system absorbs 500 J of heat and does 200 J of work on the surroundings. What is ΔU of the system? [Cards: 33, 39]

C21–C26: Atomic Structure & Periodicity

C21. Write the ground-state electron configuration of oxygen ($Z = 8$). [Cards: 54]

C22. What are the four quantum numbers for the last electron in sodium ($Z = 11$)? [Cards: 53, 54]

C23. How many orbitals are in the $3d$ subshell? How many electrons can it hold? [Cards: 58]

C24. Which is larger: a sodium atom (Na) or a sodium ion (Na^+)? Why? [Cards: 59, 63]

C25. Which has the higher ionization energy: Mg or Al ? Explain using the periodic trend. [Cards: 60]

C26. Arrange F^- , Ne , and Na^+ in order of increasing size. [Cards: 63]

C27–C32: Bonding & Molecular Structure

C27. Draw the Lewis structure of NH_3 . How many lone pairs are on the central atom? [Cards: 64]

C28. What is the formal charge on the central atom in SO_4^{2-} if each S–O bond is single? (Assume octets on all O atoms.) [Cards: 64, 65]

C29. Predict the molecular geometry of CO_2 . [Cards: 68, 69]

C30. What is the hybridization of the central atom in BH_3 ? [Cards: 68, 70]

C31. Is CO_2 a polar molecule? Explain. [Cards: 72, 73]

C32. Which has the higher lattice energy: NaCl or MgO ? Why? [Cards: 74]

C33–C38: Kinetics Basics

C33. For the reaction $A \rightarrow \text{products}$, doubling $[A]$ doubles the rate. What is the order with respect to A?

[Cards: 76, 78]

C34. A first-order reaction has $k = 0.0693 \text{ s}^{-1}$. What is its half-life? [Cards: 81, 83]

C35. The half-life of a first-order reaction is 100 s. If $[A]_0 = 2.00 \text{ M}$, what is $[A]$ after 200 s? [Cards: 81, 83]

C36. For a zero-order reaction, which plot is linear? [Cards: 80]

C37. The rate law is $\text{Rate} = k[A]^2[B]$. What is the overall reaction order? [Cards: 76, 79]

C38. After three half-lives, what fraction of a radioactive sample remains? [Cards: 81, 83, 146]

C39–C44: Equilibrium Basics

C39. Write the equilibrium expression K_c for: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$. [Cards: 89]

C40. For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, does $K_c = K_p$? Explain. [Cards: 90]

C41. If $K_c = 1.0 \times 10^3$ for a reaction, are products or reactants favored at equilibrium? [Cards: 92]

C42. A system at equilibrium has $Q = K$. If more reactant is added, what happens to Q ? Which direction does the reaction shift? [Cards: 91, 97]

C43. For the exothermic reaction $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$, does increasing temperature shift equilibrium left or right? [Cards: 96, 99]

C44. For $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, increasing pressure shifts equilibrium which way? [Cards: 96, 98]

C45–C50: Acids & Bases Basics

C45. Calculate the pH of 0.010 M HCl. (HCl is a strong acid.) [Cards: 101, 107, 189]

C46. What is the pOH of a solution with $[\text{OH}^-] = 1.0 \times 10^{-3} \text{ M}$? What is its pH? [Cards: 102, 103]

C47. A solution has pH = 4.50. What is $[\text{H}^+]$? [Cards: 101]

C48. Write the K_a expression for acetic acid (CH_3COOH). [Cards: 104]

C49. The K_a of a weak acid is 1.8×10^{-5} . What is its $\text{p}K_a$? [Cards: 104]

C50. Calculate $[\text{OH}^-]$ in a 0.20 M NaOH solution. (NaOH is a strong base.) [Cards: 108, 190]

INTERMEDIATE — 50 Problems (Level 2–3, 2–5 cards)

C51–C58: Stoichiometry — Limiting, Yield, Formulas

C51. 10.0 g of H_2 reacts with 64.0 g of O_2 to make water. Identify the limiting reagent and calculate the theoretical yield of water in grams. ($M_{\text{H}_2} = 2.00$, $M_{\text{O}_2} = 32.0$, $M_{\text{H}_2\text{O}} = 18.0$) [Cards: 1, 8, 9, 11, 12]

C52. In problem C51, the actual yield of water is 25.0 g. Calculate the percent yield. [Cards: 12, 13]

C53. 15.0 g of Al reacts with 20.0 g of Cl_2 . ($2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$, $M_{\text{Al}} = 27.0$, $M_{\text{Cl}_2} = 71.0$, $M_{\text{AlCl}_3} = 133.5$) Find the limiting reagent and the mass of AlCl_3 produced. [Cards: 1, 8, 9, 11, 12]

C54. In problem C53, how many grams of the excess reagent remain? [Cards: 11, 12, 14]

C55. A compound is 40.0% C, 6.71% H, and 53.3% O by mass. Find its empirical formula. ($M_{\text{C}} = 12.0$, $M_{\text{H}} = 1.01$, $M_{\text{O}} = 16.0$) [Cards: 15]

C56. The empirical formula from C55 has molar mass 180 g/mol. What is the molecular formula? [Cards: 15, 16]

C57. Combustion of 0.500 g of a hydrocarbon produces 1.65 g of CO_2 and 0.675 g of H_2O . Find the empirical formula. ($M_{\text{CO}_2} = 44.0$, $M_{\text{H}_2\text{O}} = 18.0$, $M_{\text{C}} = 12.0$, $M_{\text{H}} = 1.01$) [Cards: 18]

C58. A compound is found to be 92.3% C and 7.7% H by mass. Its molar mass is 78.0 g/mol. Determine the molecular formula. [Cards: 15, 16, 17]

C59–C66: Gases — Mixtures, Effusion, Real Gases

C59. A mixture of 2.00 mol N_2 and 3.00 mol O_2 is in a 10.0 L tank at 300 K. Find the total pressure and the partial pressure of N_2 . ($R = 0.08206$) [Cards: 19, 26, 27]

C60. Hydrogen gas is collected over water at 25°C. The total pressure is 750 torr. The vapor pressure of water at 25°C is 23.8 torr. What is the partial pressure of the dry hydrogen? [Cards: 28]

C61. Calculate the ratio of effusion rates of H_2 ($M = 2.02$) to O_2 ($M = 32.0$). [Cards: 29]

C62. Calculate the root-mean-square speed of N_2 molecules at 300 K. ($M_{\text{N}_2} = 0.0280$ kg/mol, $R = 8.314$ J mol⁻¹ K⁻¹) [Cards: 30]

C63. What is the average kinetic energy per mole of an ideal gas at 400 K? ($R = 8.314$ J mol⁻¹ K⁻¹) [Cards: 31]

C64. A gas mixture contains 0.500 mol He and 1.50 mol Ne. The total pressure is 2.00 atm. Find the partial pressure of He. [Cards: 26, 27]

C65. For CO_2 , $a = 3.59$ L² atm mol⁻² and $b = 0.0427$ L mol⁻¹. Write the van der Waals equation for 1.00 mol and explain what a and b correct for. [Cards: 32]

C66. A sealed 2.00 L flask at 300 K contains 0.500 mol N_2 and an unknown amount of Ar. The total pressure is 8.20 atm. How many moles of Ar are present? ($R = 0.08206$) [Cards: 19, 26]

C67–C74: Thermochemistry — Hess, Calorimetry, ΔG

C67. Calculate ΔH° for: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ using standard enthalpies of formation: $\Delta H_f^\circ(\text{CH}_4) = -74.8$, $\Delta H_f^\circ(\text{CO}_2) = -393.5$, $\Delta H_f^\circ(\text{H}_2\text{O}(\text{l})) = -285.8$ kJ/mol. [Cards: 43]

C68. Given: $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) \Delta H = -393.5$ kJ; $\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) \Delta H = -283.0$ kJ. Use Hess's Law to find ΔH for $\text{C}(\text{s}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}(\text{g})$. [Cards: 42]

C69. Estimate ΔH for $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$ using bond energies: $\text{BE}(\text{H-H}) = 436$, $\text{BE}(\text{Cl-Cl}) = 243$, $\text{BE}(\text{H-Cl}) = 432$ kJ/mol. [Cards: 44]

C70. 0.500 g of methane (CH_4) is burned in a bomb calorimeter ($C_{\text{cal}} = 12.0$ kJ/K). The temperature rises by 2.08 K. Find ΔU of combustion per mole of CH_4 . ($M_{\text{CH}_4} = 16.0$) [Cards: 37]

C71. Calculate ΔG° for a reaction at 298 K where $\Delta H^\circ = -92.2$ kJ/mol and $\Delta S^\circ = -198.8$ J mol⁻¹ K⁻¹. Is the reaction spontaneous at 298 K? [Cards: 48, 49]

C72. For the reaction in C71, at what temperature does $\Delta G^\circ = 0$? [Cards: 48, 52]

C73. Calculate ΔG° given $K = 1.8 \times 10^{-5}$ at 298 K. ($R = 8.314$ J mol⁻¹ K⁻¹) [Cards: 51]

C74. Calculate ΔS° for $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ using tabulated S° values: $S^\circ(\text{N}_2) = 191.5$, $S^\circ(\text{H}_2) = 130.6$, $S^\circ(\text{NH}_3) = 192.3$ J mol⁻¹ K⁻¹. Explain why ΔS° is negative. [Cards: 46, 47]

C75–C82: Bonding & Structure — Deeper Analysis

C75. Draw all valid resonance structures for the nitrate ion (NO_3^-). What is the average N–O bond order? [Cards: 64, 66]

C76. Assign formal charges in the Lewis structure of CO. Which atom carries the negative formal charge? [Cards: 64, 65]

C77. Predict the molecular geometry and bond angle of SF_4 . [Cards: 68, 69, 71]

C78. What is the hybridization of the central atom in PCl_5 ? [Cards: 68, 70]

C79. BeCl_2 is linear while H_2O is bent. Explain using VSEPR theory. [Cards: 68, 69]

C80. Rank NaF, MgO, and KBr in order of increasing lattice energy. Explain. [Cards: 74]

C81. The bond length of $\text{N} \equiv \text{N}$ is 110 pm; $\text{N} = \text{N}$ is 125 pm; $\text{N} - \text{N}$ is 145 pm. Explain the trend. [Cards: 67, 75]

C82. BF_3 is nonpolar despite having polar B–F bonds. Explain. [Cards: 68, 72, 73]

C83–C90: Kinetics — Integrated Rate Laws, Arrhenius, Mechanisms

C83. A reaction is first-order with $k = 0.0231$ min⁻¹. How long does it take for $[\text{A}]$ to decrease from 1.00 M to 0.250 M? [Cards: 81, 83]

C84. For a second-order reaction, $k = 0.540$ M⁻¹ s⁻¹ and $[\text{A}]_0 = 0.200$ M. Calculate $[\text{A}]$ after 30.0 s. [Cards: 82]

C85. A reaction has $k_1 = 0.0200$ s⁻¹ at 300 K and $k_2 = 0.0800$ s⁻¹ at 320 K. Calculate E_a . ($R = 8.314$ J mol⁻¹ K⁻¹) [Cards: 85]

C86. The activation energy of a reaction is 75.0 kJ/mol. By what factor does the rate increase when the temperature is raised from 300 K to 310 K? ($R = 8.314$) [Cards: 84, 85]

C87. A proposed mechanism: Step 1 (slow): $\text{A} + \text{B} \rightarrow \text{C}$. Step 2 (fast): $\text{C} + \text{A} \rightarrow \text{D}$. What is the overall reaction? What is the rate law? [Cards: 76, 87]

C88. How does a catalyst increase the rate of a reaction? Does it change ΔH , ΔG , or K ? [Cards: 86, 88]

C89. The following kinetic data were obtained for $A + B \rightarrow \text{products}$:

Trial	$[A]_0$ (M)	$[B]_0$ (M)	Initial Rate (M/s)
1	0.100	0.100	4.0×10^{-5}
2	0.200	0.100	1.6×10^{-4}
3	0.100	0.200	8.0×10^{-5}

Determine the rate law and the value of k with units. [Cards: 76, 77, 78]

C90. A zero-order reaction has $k = 0.0150 \text{ M s}^{-1}$ and $[A]_0 = 0.600 \text{ M}$. How long until the reactant is completely consumed? [Cards: 80]

C91–C98: Equilibrium — ICE Tables & Applications

C91. For $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, $K_c = 50.0$ at 700 K. Initially $[\text{H}_2] = [\text{I}_2] = 1.00 \text{ M}$, $[\text{HI}] = 0$. Set up the ICE table and write the expression to solve for equilibrium concentrations. [Cards: 89, 93, 95]

C92. For a reaction with $K_c = 1.0 \times 10^{-4}$, the initial concentrations are $[A]_0 = 0.500 \text{ M}$, $[B]_0 = 0$. Can the small- K approximation be used? If so, find $[B]$ at equilibrium for $A \rightleftharpoons 2B$. [Cards: 89, 93, 94]

C93. For $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$, $K_p = 1.05$ at 250°C . What is K_c at this temperature? ($R = 0.08206$) [Cards: 90]

C94. At a certain moment, $[\text{SO}_2] = 0.20 \text{ M}$, $[\text{O}_2] = 0.10 \text{ M}$, $[\text{SO}_3] = 0.40 \text{ M}$ for $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ with $K_c = 4.3 \times 10^2$. Calculate Q and predict the direction the reaction must shift to reach equilibrium. [Cards: 89, 91]

C95. For the exothermic reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, predict the effect of (a) adding N_2 , (b) decreasing volume, (c) increasing temperature on the equilibrium position. [Cards: 96, 97, 98, 99]

C96. 0.500 mol of HI is placed in a 1.00 L flask at 520 K. At equilibrium, $[\text{HI}] = 0.400 \text{ M}$. Calculate K_c for $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$. [Cards: 89, 93, 95]

C97. For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, $K_p = 0.148$ at 298 K. Initially $P_{\text{N}_2\text{O}_4} = 1.00 \text{ atm}$, $P_{\text{NO}_2} = 0$. Find the equilibrium partial pressures. (Solve the quadratic — do NOT use approximation.) [Cards: 89, 93, 95]

C98. $K_c = 0.212$ at 100°C for $2\text{NOBr}(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{Br}_2(\text{g})$. In a 2.00 L flask, 0.400 mol NOBr , 0.200 mol NO , and 0.300 mol Br_2 are mixed. Is the system at equilibrium? If not, which way does it shift? [Cards: 89, 91, 93]

C99–C100: Cross-Domain Bridges

C99. 5.00 g of CaCO_3 ($M = 100.1$) is heated to decompose: $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$. What volume of CO_2 is produced at STP? [Cards: 1, 6, 7, 8, 9, 160]

C100. The combustion of 2.00 g of propane (C_3H_8 , $M = 44.1$) releases 101 kJ of heat ($\Delta H_{\text{comb}} = -2220 \text{ kJ/mol}$). This heat is absorbed by 500 g of water initially at 22.0°C . Find the final temperature of

the water. ($c_w = 4.184$) [Cards: 1, 33, 34, 162]

ADVANCED — 50 Problems (Level 3–4, 4–10 cards)

C101–C110: Acids & Bases — Buffers, Titrations, Polyprotic

C101. Calculate the pH of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$). [Cards: 101, 104, 109]

C102. Calculate the pH of 0.100 M NH_3 ($K_b = 1.8 \times 10^{-5}$). [Cards: 101, 102, 105, 110]

C103. A buffer is prepared by mixing 0.500 mol of acetic acid (CH_3COOH) and 0.500 mol of sodium acetate (CH_3COONa) in 1.00 L of solution. $K_a = 1.8 \times 10^{-5}$. Calculate the pH. [Cards: 104, 111, 112, 113, 169]

C104. To the buffer in C103, 0.100 mol of HCl is added (assume no volume change). Calculate the new pH. [Cards: 104, 111, 112, 169]

C105. A 25.0 mL sample of 0.100 M HCl is titrated with 0.100 M NaOH. Calculate the pH after adding (a) 0 mL, (b) 12.5 mL, (c) 25.0 mL, (d) 30.0 mL of NaOH. [Cards: 101, 107, 114]

C106. A 50.0 mL sample of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) is titrated with 0.100 M NaOH. Calculate the pH at (a) 0 mL, (b) 25.0 mL (half-equivalence), (c) 50.0 mL (equivalence) of NaOH added. [Cards: 101, 104, 109, 112, 114, 115, 116, 170]

C107. 0.200 M sodium acetate (CH_3COONa) dissolves in water. Calculate the pH of the solution. (K_a of $\text{CH}_3\text{COOH} = 1.8 \times 10^{-5}$) [Cards: 101, 104, 105, 119]

C108. Calculate the pH of 0.100 M H_2CO_3 (a diprotic acid with $K_{a1} = 4.3 \times 10^{-7}$, $K_{a2} = 5.6 \times 10^{-11}$). [Cards: 101, 104, 109, 118]

C109. What is the percent ionization of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$)? What is the percent ionization if the solution is diluted to 0.0100 M? [Cards: 101, 104, 106, 109]

C110. A buffer is needed at pH 9.25. Which acid/conjugate-base pair is best: (a) $\text{CH}_3\text{COOH}/\text{CH}_3\text{COO}^-$ ($\text{p}K_a = 4.74$), (b) $\text{NH}_4^+/\text{NH}_3$ ($\text{p}K_a = 9.25$), (c) $\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$ ($\text{p}K_a = 7.21$)? What ratio of $[\text{A}^-]/[\text{HA}]$ is needed? [Cards: 104, 112, 113, 169]

C111–C120: Electrochemistry — Cells, Nernst, Electrolysis

C111. For the reaction $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$, given $E_{\text{Zn}^{2+}/\text{Zn}}^\circ = -0.76 \text{ V}$ and $E_{\text{Cu}^{2+}/\text{Cu}}^\circ = +0.34 \text{ V}$, calculate E_{cell}° . Is the reaction spontaneous? [Cards: 121, 122, 125, 126]

C112. For the cell in C111, calculate ΔG° and K at 298 K. ($F = 96,485 \text{ C/mol e}^-$, $R = 8.314$) [Cards: 51, 125, 127, 129]

C113. Calculate the cell potential at 298 K for: $\text{Zn(s)}|\text{Zn}^{2+}(0.010 \text{ M})||\text{Cu}^{2+}(1.00 \text{ M})|\text{Cu(s)}$. $E_{\text{cell}}^\circ = 1.10 \text{ V}$. [Cards: 125, 128]

C114. A current of **2.50 A** is passed through molten **NaCl** for **30.0 minutes**. What mass of sodium metal is produced? ($M_{\text{Na}} = 23.0$, $F = 96,485$) [Cards: 129, 131, 132]

C115. Balance the following redox reaction in acidic solution: $\text{MnO}_4^- + \text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + \text{Fe}^{3+}$. [Cards: 121, 122, 123, 124]

C116. For the reaction $2\text{Al}(\text{s}) + 3\text{I}_2(\text{s}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 6\text{I}^{-}(\text{aq})$, $E_{\text{cell}}^{\circ} = 2.20 \text{ V}$. Calculate ΔG° and determine K at **298 K**. [Cards: 51, 125, 127, 129, 166]

C117. An electrolytic cell deposits **1.50 g** of **Cu** from **CuSO₄** solution using a current of **1.80 A**. How long did the electrolysis take? ($M_{\text{Cu}} = 63.5$, $F = 96,485$) [Cards: 129, 131, 132]

C118. Calculate E_{cell}° for: $2\text{Fe}^{3+}(\text{aq}) + 2\text{I}^{-}(\text{aq}) \rightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{I}_2(\text{s})$. Given: $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^{\circ} = +0.77 \text{ V}$, $E_{\text{I}_2/\text{I}^{-}}^{\circ} = +0.54 \text{ V}$. Is this reaction spontaneous under standard conditions? [Cards: 121, 122, 125, 126]

C119. For the cell $\text{Pt}|\text{H}_2(1 \text{ atm})|\text{H}^{+}(\text{pH} = ?)||\text{Cu}^{2+}(1.00 \text{ M})|\text{Cu}(\text{s})$, the measured potential is **0.480 V**. Given $E_{\text{cell}}^{\circ} = 0.34 \text{ V}$, find the pH of the anode compartment. [Cards: 101, 125, 128]

C120. A galvanic cell uses the reaction $\text{Cd}(\text{s}) + \text{Ni}^{2+}(\text{aq}) \rightarrow \text{Cd}^{2+}(\text{aq}) + \text{Ni}(\text{s})$ with $E_{\text{cell}}^{\circ} = 0.15 \text{ V}$. Initially $[\text{Ni}^{2+}] = 1.00 \text{ M}$, $[\text{Cd}^{2+}] = 0.0100 \text{ M}$. Calculate the initial cell potential and predict whether it will increase or decrease as the cell discharges. [Cards: 125, 126, 128]

C121–C130: Solutions — Colligative Properties & Concentration

C121. Calculate the molality of a solution made by dissolving **10.0 g** of **NaCl** ($M = 58.5$) in **200 g** of water. [Cards: 134]

C122. What is the boiling point of an aqueous solution containing **34.2 g** of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$, $M = 342$) in **500 g** of water? ($K_b = 0.512 \text{ K kg mol}^{-1}$, $i = 1$ for sucrose) [Cards: 134, 141, 144]

C123. Calculate the freezing point of a solution of **5.85 g** of **NaCl** ($M = 58.5$) dissolved in **100 g** of water. ($K_f = 1.86 \text{ K kg mol}^{-1}$) [Cards: 134, 142, 144]

C124. What is the osmotic pressure at **298 K** of a **0.0100 M** aqueous solution of **CaCl₂** (assume complete dissociation)? ($R = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$) [Cards: 143, 144]

C125. The vapor pressure of pure water at **25°C** is **23.8 torr**. What is the vapor pressure of a solution containing **18.0 g** of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$, $M = 180$) in **90.0 g** of water? ($M_{\text{H}_2\text{O}} = 18.0$) [Cards: 136, 140]

C126. What concentration of **NaCl** (in mol/L) is isotonic with blood, which has an osmotic pressure of **7.7 atm** at **310 K**? ($R = 0.08206$, assume $i = 2$) [Cards: 143, 144]

C127. **25.0 mL** of **6.00 M HCl** is diluted to **500 mL**. What is the new concentration? [Cards: 133, 137]

C128. A solution is prepared by mixing **50.0 g** of ethanol ($\text{C}_2\text{H}_5\text{OH}$, $M = 46.1$) with **50.0 g** of water ($M = 18.0$). Calculate the mole fraction of each component. [Cards: 136]

C129. Rank the following aqueous solutions in order of increasing freezing point: **0.10 m NaCl**, **0.10 m CaCl₂**, **0.10 m C₆H₁₂O₆**, **0.10 m CH₃COOH** (weak acid, assume minimal ionization). [Cards: 139, 142, 144]

C130. A 0.500 g protein is dissolved in enough water to make 50.0 mL of solution. The osmotic pressure is 3.85 torr at 298 K. What is the molar mass of the protein? ($R = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$, $1 \text{ atm} = 760 \text{ torr}$) [Cards: 133, 143, 144, 199]

C131–C140: Nuclear Chemistry

C131. The half-life of ^{14}C is 5730 years. A sample contains 25% of the original ^{14}C . How old is the sample? [Cards: 145, 146]

C132. ^{238}U decays by alpha emission. Write the balanced nuclear equation and identify the daughter nuclide. [Cards: 148]

C133. ^{131}I ($t_{1/2} = 8.02 \text{ days}$) is used in medical imaging. What fraction remains after 30.0 days? [Cards: 145, 146]

C134. Calculate the binding energy per nucleon of ^{56}Fe (mass = 55.9349 u). ($m_p = 1.00728 \text{ u}$, $m_n = 1.00867 \text{ u}$, $1 \text{ u} = 931.5 \text{ MeV}/c^2$) [Cards: 153, 154, 155, 207]

C135. ^{60}Co decays by β^- emission with $t_{1/2} = 5.27 \text{ years}$. A sample has an activity of $2.50 \times 10^4 \text{ Bq}$. What was the activity 10.0 years ago? [Cards: 145, 146, 147, 149]

C136. Complete the nuclear equation: $^{238}_{92}\text{U} \rightarrow ^{234}_{90}\text{Th} + ?$. What type of decay is this? [Cards: 148]

C137. A rock contains 0.500 g of ^{238}U ($t_{1/2} = 4.47 \times 10^9 \text{ yr}$) and 0.125 g of ^{206}Pb . Assuming all the lead came from uranium decay, estimate the age of the rock. ($M_{238} = 238$, $M_{206} = 206$) [Cards: 1, 145, 146]

C138. Calculate the mass defect and binding energy of ^4He (mass = 4.00260 u). ($m_p = 1.00728 \text{ u}$, $m_n = 1.00867 \text{ u}$, $1 \text{ u} = 931.5 \text{ MeV}/c^2$) [Cards: 153, 154, 207]

C139. The decay constant of ^{32}P is 0.0485 day^{-1} . Calculate its half-life and the time for 90% to decay. [Cards: 145, 146]

C140. ^{235}U ($t_{1/2} = 7.04 \times 10^8 \text{ yr}$) and ^{238}U ($t_{1/2} = 4.47 \times 10^9 \text{ yr}$) are both present in natural uranium. Today, ^{235}U is 0.72% of natural uranium. If the isotopes were originally equally abundant, how long ago did they form? (Solve using the ratio of decay equations.) [Cards: 145, 146]

C141–C150: Multi-Domain Synthesis — Olympiad Level

C141. 10.0 g of impure CaCO_3 ($M = 100.1$) is treated with excess HCl. The CO_2 produced is collected over water at 25°C ($P_{\text{H}_2\text{O}} = 23.8 \text{ torr}$). The volume of gas collected is 2.15 L at a total pressure of 755 torr. What is the percent purity of the CaCO_3 ? ($R = 0.08206$) [Cards: 1, 8, 9, 19, 20, 28, 160]

C142. 0.500 mol of N_2O_4 is placed in a 5.00 L flask at 298 K. At equilibrium, the total pressure is 3.20 atm. Calculate K_p and K_c for $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$. ($R = 0.08206$) [Cards: 19, 89, 90, 93, 95]

C143. A 0.250 M solution of a weak monoprotic acid has pH = 4.25. Calculate K_a and the percent ionization. What is the freezing point depression of this solution? ($K_f = 1.86 \text{ K kg mol}^{-1}$, assume density $\approx 1.00 \text{ g/mL}$, neglect ion pairing.) [Cards: 101, 104, 106, 109, 134, 142, 144]

C144. The reaction $\text{A} \rightarrow \text{products}$ is first-order. At 300 K, $k = 0.0200 \text{ min}^{-1}$. At 320 K, $k = 0.0600 \text{ min}^{-1}$. Calculate E_a and predict k at 340 K. If $[\text{A}]_0 = 1.50 \text{ M}$, how long at 340 K to reach

0.150 M? [Cards: 81, 83, 84, 85, 86]

C145. A voltaic cell is constructed: $\text{Mg(s)}|\text{Mg}^{2+}(0.050\text{ M})||\text{Ag}^+(0.010\text{ M})|\text{Ag(s)}$. $E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.37\text{ V}$, $E^\circ_{\text{Ag}^+/\text{Ag}} = +0.80\text{ V}$. Calculate E°_{cell} , E_{cell} at the given concentrations, ΔG , and K at 298 K. ($F = 96,485$, $R = 8.314$) [Cards: 51, 125, 126, 127, 128, 165, 166]

C146. 2.00 g of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$, $M = 122$) is dissolved in 50.0 g of benzene. The freezing point is lowered by 1.62 K. K_f for benzene is $5.12\text{ K kg mol}^{-1}$. Benzoic acid dimerizes in benzene: $2\text{C}_6\text{H}_5\text{COOH} \rightleftharpoons (\text{C}_6\text{H}_5\text{COOH})_2$. Calculate the equilibrium constant K_c for the dimerization. (Assume 1 L of benzene solution $\approx 1\text{ kg}$ for concentration calculation.) [Cards: 89, 93, 95, 134, 142, 144]

C147. For the reaction $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$, the rate law is $\text{Rate} = k[\text{NO}]^2[\text{O}_2]$. At 298 K, $k = 7.1 \times 10^3\text{ M}^{-2}\text{ s}^{-1}$. An industrial reactor initially has $P_{\text{NO}} = 0.500\text{ atm}$ and $P_{\text{O}_2} = 0.250\text{ atm}$ at 400 K in a 10.0 L vessel. Find the initial rate in M/s. ($R = 0.08206$) [Cards: 19, 76, 77, 133]

C148. 1.00 L of a buffer contains 0.200 mol CH_3COOH and 0.300 mol CH_3COONa . ($K_a = 1.8 \times 10^{-5}$) (a) Calculate the initial pH. (b) 0.0500 mol of NaOH is added. Calculate the new pH. (c) 0.0500 mol of HCl is added (to the original buffer). Calculate the new pH. (d) Compare the pH changes — is the buffer effective? [Cards: 101, 104, 111, 112, 113, 169]

C149. The solubility product of AgCl is $K_{sp} = 1.8 \times 10^{-10}$. Ag^+ forms a complex with NH_3 : $\text{Ag}^+ + 2\text{NH}_3 \rightleftharpoons \text{Ag}(\text{NH}_3)_2^+$ with $K_f = 1.7 \times 10^7$. Calculate the molar solubility of AgCl in 1.0 M NH_3 . (Hint: Combine the two equilibria; the very large K_f means you may assume nearly complete complexation.) [Cards: 89, 93, 95]

C150. Consider the reaction $\text{A(g)} \rightleftharpoons \text{B(g)} + \text{C(g)}$ in a 2.00 L flask at 500 K. Initially $P_A = 1.50\text{ atm}$. At equilibrium, the total pressure is 2.10 atm. (a) Calculate K_p . (b) ΔG° for this reaction at 500 K. (c) If $\Delta H^\circ = 42.0\text{ kJ/mol}$ (constant over temperature), calculate K_p at 600 K. (d) Predict the equilibrium shift if the volume is halved. ($R = 8.314\text{ J mol}^{-1}\text{ K}^{-1}$) [Cards: 19, 48, 51, 89, 90, 93, 95, 96, 98, 99, 164]

INDUSTRIAL & APPLIED CHEMISTRY — 30 Problems (Level 2–4, 2–8 cards)

Real processes. Real numbers. Same cards. The Haber plant, the chlor-alkali cell, the oil refinery — they all obey the same equations you already know. The only difference is that the "given" values sound like an engineer's clipboard instead of a textbook.

C151–C158: Haber-Bosch Process — Ammonia Synthesis

C151. The Haber process: $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons 2\text{NH}_3\text{(g)}$, $\Delta H^\circ = -92.2\text{ kJ/mol}$. A reactor at 450°C and 200 atm uses a Fe/Ru catalyst. (a) Explain why high pressure favors ammonia. (b) Explain why the reaction is run at 450°C despite being exothermic. (c) What does the catalyst do — and what does it NOT do? [Cards: 40, 86, 88, 96, 98, 99]

C152. A Haber reactor is fed 100 mol/min of N_2 and 300 mol/min of H_2 . At equilibrium at 400°C, the mole fraction of NH_3 in the exit stream is 0.25. What is the single-pass conversion of N_2 ? [Cards: 8, 9, 93, 95]

C153. In the Haber process, $\Delta G^\circ = -33.0 \text{ kJ/mol}$ at 298 K (see C71). At the operating temperature of 723 K, $\Delta G^\circ \approx +20 \text{ kJ/mol}$ (nonspontaneous under standard conditions). Explain why the industrial process still works. Calculate K_p at 723 K from ΔG° . ($R = 8.314$) [Cards: 48, 49, 51, 99]

C154. A Haber plant produces 1000 metric tons of NH_3 per day (1 ton = 10^6 g). What mass of N_2 is consumed daily, assuming 95% overall yield (including recycle)? ($M_{\text{NH}_3} = 17.0$, $M_{\text{N}_2} = 28.0$) [Cards: 1, 8, 9, 10, 13]

C155. The activation energy of ammonia synthesis on an iron catalyst is 160 kJ/mol (uncatalyzed: 940 kJ/mol). By what factor does the catalyst increase the rate at 723 K? ($R = 8.314$) [Cards: 84, 85, 86, 88]

C156. ΔS° for the Haber reaction is $-198.8 \text{ J mol}^{-1} \text{ K}^{-1}$ (see C74). The plant engineer notes that at 723 K, the ΔG° calculated from $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ is +20 kJ/mol. Verify this calculation given $\Delta H^\circ = -92.2 \text{ kJ/mol}$. Why does the entropy term dominate at high T ? [Cards: 46, 48, 52]

C157. The feed to a Haber reactor is a 3 : 1 H_2 : N_2 mixture at 500 K and 300 atm total pressure. At equilibrium, the partial pressure of NH_3 is 60.0 atm. Calculate K_p . (Hint: use Dalton's Law to find P_{N_2} and P_{H_2} .) [Cards: 19, 26, 27, 89, 93, 95]

C158. The Haber process uses a recycle loop: unreacted N_2 and H_2 are separated from the ammonia product and returned to the reactor. If the single-pass conversion of N_2 is 15%, how many passes through the reactor are needed (theoretically) for 99% overall conversion? (Treat each pass as independent; use $(1 - 0.15)^n = 0.01$.) [Cards: 81, 145]

C159–C164: Contact Process — Sulfuric Acid

C159. The Contact process step: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H^\circ = -197.8 \text{ kJ/mol}$. At 450°C, $K_p = 4.3 \times 10^2$. A reactor initially has $P_{\text{SO}_2} = 1.00 \text{ atm}$, $P_{\text{O}_2} = 0.50 \text{ atm}$, $P_{\text{SO}_3} = 0$. Calculate the equilibrium partial pressures of all species. [Cards: 89, 93, 95]

C160. In the Contact process, the SO_2 oxidation is run at 450°C with a V_2O_5 catalyst. At 298 K, $K_p \approx 10^{25}$, but the reaction is kinetically frozen (rates are negligible). Explain the trade-off between thermodynamics (high K at low T) and kinetics (reasonable rate only at high T). [Cards: 40, 84, 86, 88, 92, 99]

C161. Roasting of pyrite (FeS_2) produces SO_2 for the Contact process:
 $4\text{FeS}_2 + 11\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$. A roaster processes 500 kg of pyrite (assume 100% FeS_2 , $M = 120$) per hour. What volume of SO_2 at STP is produced per hour? [Cards: 1, 6, 7, 8, 9, 160]

C162. The SO_3 from the Contact process is absorbed in 98% H_2SO_4 (not water — to avoid acid mist). The absorption reaction: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ (oleum), then $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4$. If a plant absorbs 1000 kg of SO_3 ($M = 80.1$) per hour, what mass of 100% H_2SO_4 ($M = 98.1$) is ultimately produced? (The net is $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$.) [Cards: 1, 8, 9, 10]

C163. V_2O_5 catalyzes SO_2 oxidation via a two-step redox mechanism: $\text{V}_2\text{O}_5 + \text{SO}_2 \rightarrow \text{V}_2\text{O}_4 + \text{SO}_3$, then $\text{V}_2\text{O}_4 + \frac{1}{2}\text{O}_2 \rightarrow \text{V}_2\text{O}_5$. (a) What is the net reaction? (b) Why is V_2O_5 a catalyst rather than a reactant? (c) Assign oxidation numbers to V in V_2O_5 and V_2O_4 . [Cards: 88, 121, 122, 123]

C164. An Contact process plant operates at 98% conversion of SO_2 to SO_3 , and 99.5% absorption of SO_3 into H_2SO_4 . If the plant feeds 200 kg of SO_2 ($M = 64.1$) per hour, what is the daily production of H_2SO_4 ($M = 98.1$) in metric tons? Assume 24-hour operation. [Cards: 1, 8, 9, 10, 13]

C165–C170: Chlor-Alkali & Electrochemical Industry

C165. The chlor-alkali process electrolyzes brine (NaCl solution):

$2\text{NaCl}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq}) + \text{Cl}_2(\text{g}) + \text{H}_2(\text{g})$. A membrane cell operates at 4.0 V with 150,000 A current. What mass of NaOH ($M = 40.0$) is produced per hour? ($F = 96,485$) [Cards: 121, 122, 123, 129, 131, 132]

C166. The chlor-alkali cell in C165 has a current efficiency of 96% (some current is lost to side reactions). What is the actual daily production of Cl_2 ($M = 71.0$) in metric tons? The half-reaction is $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$. [Cards: 129, 131, 132]

C167. Aluminum is produced by the Hall-Héroult process: Al_2O_3 (dissolved in cryolite) is electrolyzed. The half-reaction at the cathode: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$. A smelter operates at 250,000 A. How many kg of Al ($M = 27.0$) are produced per cell per day? ($F = 96,485$) [Cards: 129, 131, 132]

C168. In the Hall-Héroult process, the anode reaction consumes the carbon anode:

$\text{C}(\text{s}) + 2\text{O}^{2-} \rightarrow \text{CO}_2(\text{g}) + 4\text{e}^-$. For every 1.00 kg of Al produced, what mass of carbon ($M = 12.0$) is consumed? ($M_{\text{Al}} = 27.0$) [Cards: 1, 8, 9, 10, 131]

C169. A lithium-ion battery cell has a nominal voltage of 3.7 V and capacity of 3000 mAh (milliamp-hours).

(a) Convert the capacity to coulombs. (b) If the cell reaction transfers 1 mole of electrons per mole of Li^+ , how many grams of lithium ($M = 6.94$) are theoretically moved between electrodes during a full discharge? ($F = 96,485$) [Cards: 129, 131, 132]

C170. Electroplating: a factory plates chromium onto steel bumpers from a CrO_3 bath (Cr is reduced from Cr^{6+} to Cr^0). A bumper requires 2.00 g of Cr ($M = 52.0$) deposited. The plating bath operates at 500 A. How many seconds per bumper? ($F = 96,485$) [Cards: 121, 122, 129, 131, 132]

C171–C176: Industrial Thermochemistry & Fuels

C171. A coal-fired power plant burns C (coal approximated as pure carbon) at a rate of

2500 metric tons/day. $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$, $\Delta H^\circ = -393.5 \text{ kJ/mol}$. What is the daily heat output in GJ? ($M_{\text{C}} = 12.0$, $1 \text{ GJ} = 10^9 \text{ J}$) [Cards: 1, 39, 40]

C172. The CO_2 from C171 is captured using an amine scrubber that absorbs 90% of the CO_2 . What mass of CO_2 ($M = 44.0$) is captured per day? What volume would this CO_2 occupy at 25°C and 1.00 atm? ($R = 0.08206$) [Cards: 1, 8, 9, 10, 19]

C173. Lime is produced by thermal decomposition of limestone: $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, $\Delta H^\circ = +178 \text{ kJ/mol}$. A lime kiln fires natural gas (CH_4 , $\Delta H_{\text{comb}} = -890 \text{ kJ/mol}$) to supply the heat. How many kg of CH_4 ($M = 16.0$) are needed to decompose 1000 kg of CaCO_3 ($M = 100.1$), assuming 60% thermal efficiency? [Cards: 1, 8, 9, 10, 39, 40, 41, 162]

C174. Syngas ($\text{CO} + \text{H}_2$) is produced by steam reforming: $\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$, $\Delta H^\circ = +206 \text{ kJ/mol}$. The reaction is run at 800°C and 25 atm. (a) Why high temperature? (b) Why is excess steam (a reactant) used despite the high T already favoring products? (c) If the reactor produces

1000 m³ of syngas (measured at STP) per hour, how many moles of H₂ are in this stream if the CO : H₂ ratio is 1 : 3? [Cards: 6, 7, 19, 26, 39, 41, 96, 97, 99]

C175. Water-gas shift reaction: $\text{CO(g)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$, $\Delta H^\circ = -41.2 \text{ kJ/mol}$. At 500 K, $K_p = 126$. In an industrial shift reactor at 500 K, the inlet gas contains 1.00 mol CO, 3.00 mol H₂O, 0.50 mol CO₂, and 0.50 mol H₂ (per unit volume). (a) Calculate Q . (b) Which direction does the reaction proceed? (c) At equilibrium, 99% of CO has reacted. Verify this with an ICE table. [Cards: 89, 91, 93, 95]

C176. Methanol synthesis: $\text{CO(g)} + 2\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_3\text{OH(g)}$, $\Delta H^\circ = -90.7 \text{ kJ/mol}$. At 500 K, $K_p = 6.1 \times 10^{-3}$. Why is this reaction run at high pressure (50–100 atm) despite the catalyst (Cu/ZnO) working best at only 250°C? Use Le Chatelier's principle and the Arrhenius concept. [Cards: 40, 84, 86, 88, 96, 98, 99]

C177–C180: Polymer, Environmental & Materials Industry

C177. Polyethylene is produced by polymerization of ethylene: $n\text{C}_2\text{H}_4 \rightarrow (\text{C}_2\text{H}_4)_n$. If a reactor polymerizes 1000 kg of ethylene ($M = 28.1$) at 98% conversion, what mass of polyethylene is produced? (Polymerization is addition — all atoms of the monomer end up in the polymer.) The polyethylene has an average molecular weight of 280,000 g/mol. What is the average value of n ? [Cards: 1, 13, 16]

C178. Reverse osmosis desalination: seawater (0.60 M NaCl at 298 K) is pressurized against a semipermeable membrane. (a) Calculate the minimum pressure (osmotic pressure) that must be overcome. ($R = 0.08206$, $i = 2$) (b) In practice, a pressure of 50 atm is applied. What is the net driving pressure? (c) If the plant produces 1000 m³ of fresh water per day, at an energy cost of 4.0 kWh/m³, what is the daily energy consumption in GJ? (1 kWh = $3.6 \times 10^6 \text{ J}$) [Cards: 143, 144]

C179. Flue gas desulfurization: SO₂ is scrubbed from power plant exhaust with limestone slurry: $\text{CaCO}_3\text{(s)} + \text{SO}_2\text{(g)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CaSO}_4\text{(s)} + \text{CO}_2\text{(g)}$. A plant emits 500 kg of SO₂ ($M = 64.1$) per hour and must remove 95%. What mass of CaCO₃ ($M = 100.1$) is consumed per day? (24-hour operation) [Cards: 1, 8, 9, 10, 13]

C180. Cement production: the key reaction in a cement kiln is $\text{CaCO}_3\text{(s)} + \text{SiO}_2\text{(s)} \rightarrow \text{CaSiO}_3\text{(s)} + \text{CO}_2\text{(g)}$, $\Delta H^\circ = +88 \text{ kJ/mol}$. (a) 1000 kg of CaCO₃ is heated with stoichiometric SiO₂. What mass of CO₂ is released? ($M_{\text{CaCO}_3} = 100.1$, $M_{\text{CO}_2} = 44.0$) (b) If the heat is supplied by burning coal (approximated as C → CO₂, $\Delta H^\circ = -393.5 \text{ kJ/mol}$) with 50% thermal efficiency, what mass of carbon is needed? ($M_{\text{C}} = 12.0$) (c) Cement production accounts for ~8% of global CO₂ emissions. Using your answer to (a) and (b), calculate the total CO₂ released (process + fuel) per ton of CaCO₃ processed. [Cards: 1, 8, 9, 10, 39, 40, 41, 44, 162]

SEMICONDUCTOR PROCESS CHEMISTRY — 30 Problems (Level 2–4, 2–7 cards)

The cleanroom is a chemistry lab. Every chip layer — grown, deposited, etched, doped — is a chemical reaction controlled by the same cards you already know. The only new vocabulary is the process names. The equations are unchanged.

C181–C186: Silicon Purification & Crystal Growth

C181. Metallurgical-grade silicon is converted to trichlorosilane: $\text{Si(s)} + 3\text{HCl(g)} \rightarrow \text{SiHCl}_3\text{(g)} + \text{H}_2\text{(g)}$. A reactor treats **100 kg** of Si ($M = 28.1$) with excess HCl. What mass of SiHCl_3 ($M = 135.5$) is produced at **90%** yield? [Cards: 1, 8, 9, 10, 13]

C182. The Siemens process purifies Si by decomposing SiHCl_3 onto a heated rod:
 $2\text{SiHCl}_3\text{(g)} + 2\text{H}_2\text{(g)} \rightarrow 2\text{Si(s)} + 6\text{HCl(g)}$ (simplified). Or: $\text{SiHCl}_3\text{(g)} + \text{H}_2\text{(g)} \rightarrow \text{Si(s)} + 3\text{HCl(g)}$. A Siemens reactor deposits **5.00 kg** of ultra-pure Si ($M = 28.1$) per rod. What volume of HCl gas ($M = 36.5$) is produced at **900°C** and **1.00 atm** per rod? ($R = 0.08206$) [Cards: 1, 8, 9, 10, 19, 160]

C183. Silane (SiH_4) is an alternative silicon precursor, decomposed by CVD: $\text{SiH}_4\text{(g)} \rightarrow \text{Si(s)} + 2\text{H}_2\text{(g)}$. The decomposition is first-order with $k = 0.15 \text{ s}^{-1}$ at **650°C**. What fraction of SiH_4 decomposes in **10.0 s**? [Cards: 81, 83]

C184. Polysilicon production via the Siemens process operates at **1100°C**. At this temperature, the equilibrium constant for $\text{SiHCl}_3\text{(g)} + \text{H}_2\text{(g)} \rightleftharpoons \text{Si(s)} + 3\text{HCl(g)}$ is $K = 0.85$ (in terms of gas-phase species). If a reactor initially contains $P_{\text{SiHCl}_3} = 2.00 \text{ atm}$ and $P_{\text{H}_2} = 3.00 \text{ atm}$ (with $P_{\text{HCl}} = 0$), what is the equilibrium partial pressure of HCl? (Note: solid Si does not appear in K — Card 89.) The reaction stoichiometry gives the ICE changes: $\text{SiHCl}_3: -x$, $\text{H}_2: -x$, $\text{HCl}: +3x$. [Cards: 89, 93, 95]

C185. The Czochralski process melts polysilicon at **1414°C**. The latent heat of fusion of Si is **50.2 kJ/mol**. How much energy is required to melt **50.0 kg** of Si ($M = 28.1$) starting from solid at **25°C**? The specific heat of solid Si is **0.705 J g⁻¹ K⁻¹**. (Heat to melting point + latent heat.) [Cards: 1, 33, 39]

C186. During Czochralski crystal growth, a small amount of dopant (e.g., boron) is added to the melt. If **0.0100 g** of boron ($M = 10.8$) is added to **50.0 kg** of Si ($M = 28.1$), what is the boron concentration in (a) mass ppb, (b) mole fraction? [Cards: 17, 136, 138]

C187–C192: Thermal Oxidation & CVD

C187. Thermal oxidation of silicon: $\text{Si(s)} + \text{O}_2\text{(g)} \rightarrow \text{SiO}_2\text{(s)}$. A **200 mm** diameter wafer is oxidized until an SiO_2 layer of thickness **100 nm** (**1 nm = 10⁻⁹ m**) grows. The density of SiO_2 is **2.20 g/cm³**, $M_{\text{SiO}_2} = 60.1$. What mass of Si ($M = 28.1$) is consumed? (Volume of SiO_2 = wafer area $\times$ thickness. Wafer area = πr^2 . The mole ratio Si: SiO_2 is 1:1.) [Cards: 1, 8, 9, 10, 158]

C188. Dry oxidation of Si follows the Deal-Grove model. In the linear regime (thin oxides), the growth rate is proportional to the O_2 concentration at the surface. If the rate doubles when the O_2 partial pressure is doubled (at constant T), what does this imply about the reaction order with respect to O_2 ? Which card applies? [Cards: 76, 78]

C189. In a CVD reactor, SiO_2 is deposited from SiH_4 and O_2 : $\text{SiH}_4\text{(g)} + \text{O}_2\text{(g)} \rightarrow \text{SiO}_2\text{(s)} + 2\text{H}_2\text{(g)}$. The reactor is at **400°C** and **0.500 torr** total pressure. SiH_4 and O_2 are fed in stoichiometric ratio. What volume of SiH_4 gas (measured at STP) is consumed to deposit **1.00 g** of SiO_2 ($M = 60.1$)? [Cards: 1, 6, 7, 8, 9, 10, 160]

C190. Silicon nitride (Si_3N_4) is deposited by LPCVD:
 $3\text{SiH}_2\text{Cl}_2\text{(g)} + 4\text{NH}_3\text{(g)} \rightarrow \text{Si}_3\text{N}_4\text{(s)} + 6\text{HCl(g)} + 6\text{H}_2\text{(g)}$. (DCS = dichlorosilane, SiH_2Cl_2 .) A furnace processes **50** wafers simultaneously. Each wafer receives a **200 nm** thick Si_3N_4 film (density **3.44 g/cm³**, $M_{\text{Si}_3\text{N}_4} = 140.3$). Wafer diameter = **200 mm**. What total mass of SiH_2Cl_2 ($M = 101.0$) is consumed? [Cards: 1, 8, 9, 10, 158]

C191. Tungsten CVD uses WF_6 reduction by H_2 : $\text{WF}_6(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow \text{W}(\text{s}) + 6\text{HF}(\text{g})$. The reaction is run at 400°C and 0.80 atm total pressure. If the feed gas is 20% WF_6 and 80% H_2 by volume, what is the partial pressure of HF after 50% of the WF_6 has reacted? (Assume constant total pressure and temperature.) [Cards: 19, 26, 27]

C192. Atomic Layer Deposition (ALD) of Al_2O_3 uses alternating pulses of trimethylaluminum ($\text{Al}(\text{CH}_3)_3$) and water: $2\text{Al}(\text{CH}_3)_3(\text{g}) + 3\text{H}_2\text{O}(\text{g}) \rightarrow \text{Al}_2\text{O}_3(\text{s}) + 6\text{CH}_4(\text{g})$. Each ALD cycle deposits 0.10 nm of Al_2O_3 (density 3.95 g/cm^3 , $M_{\text{Al}_2\text{O}_3} = 102.0$). How many cycles are needed to deposit a 10.0 nm film? What mass of $\text{Al}(\text{CH}_3)_3$ ($M = 72.1$) is consumed per cm^2 of wafer area for the full 10.0 nm film? [Cards: 1, 8, 9, 10, 158]

C193–C198: Etching & Cleaning

C193. Wet etching of SiO_2 uses buffered HF : $\text{SiO}_2(\text{s}) + 6\text{HF}(\text{aq}) \rightarrow \text{H}_2\text{SiF}_6(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$. In a semiconductor fab, a 200 mm wafer with a 500 nm thick SiO_2 layer (density 2.20 g/cm^3) is completely etched. What volume of 49% HF solution (by mass, density 1.18 g/mL) is required? ($M_{\text{HF}} = 20.0$, $M_{\text{SiO}_2} = 60.1$) [Cards: 1, 8, 9, 10, 133, 135, 158]

C194. Plasma etching of silicon uses SF_6 : $2\text{SF}_6(\text{g}) \rightarrow 2\text{SF}_4(\text{g}) + 2\text{F}(\text{g})$ (simplified; F atoms then etch Si : $\text{Si} + 4\text{F} \rightarrow \text{SiF}_4$). A plasma etcher consumes 0.500 g of Si ($M = 28.1$) per minute from a wafer. What mass of SF_6 ($M = 146.1$) must be supplied per hour? (Assume 100% F utilization.) [Cards: 1, 8, 9, 10]

C195. In reactive ion etching (RIE), the etch rate of SiO_2 in a CF_4/O_2 plasma is 50 nm/min . The selectivity (ratio of etch rates) of SiO_2 to Si is $20 : 1$. After etching through a 200 nm SiO_2 layer, how much underlying Si (in nm) is unintentionally etched during a 20% over-etch (etching continues for 20% longer than the calculated endpoint time)? [Cards: — (arithmetic/applied)]

C196. RCA clean Step 1 (Standard Clean 1) uses $\text{NH}_4\text{OH}/\text{H}_2\text{O}_2/\text{H}_2\text{O}$ ($1 : 1 : 5$ ratio) at 75°C to remove organic contaminants. The solution is prepared by mixing 100 mL of 29% NH_3 (density 0.90 g/mL , $M_{\text{NH}_3} = 17.0$), 100 mL of 30% H_2O_2 (density 1.11 g/mL , $M_{\text{H}_2\text{O}_2} = 34.0$), and 500 mL of water. What is the molarity of NH_3 and H_2O_2 in the final solution? [Cards: 133, 137]

C197. After RCA cleaning, wafers are rinsed in deionized water until the resistivity of the rinse water reaches $18.2\text{ M}\Omega\text{ cm}$ (the theoretical limit for pure water). The conductivity κ is the reciprocal of resistivity: $\kappa = 1/\rho$. For pure water, $\kappa_w = 0.055\text{ }\mu\text{S/cm}$ ($\text{S} = \Omega^{-1}$). From $K_w = 1.0 \times 10^{-14}$, calculate the theoretical $[\text{H}^+]$ and $[\text{OH}^-]$ in pure water at 25°C , and verify that pure water has essentially zero ionic contaminants. [Cards: 101, 103]

C198. Piranha etch ($\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2$, $3 : 1$) cleans organic residues from wafers at 120°C . The solution is prepared by carefully adding 300 mL of 98% H_2SO_4 (density 1.84 g/mL , $M = 98.1$) to 100 mL of 30% H_2O_2 (density 1.11 g/mL , $M = 34.0$). What is the mole fraction of H_2SO_4 in the final mixture? [Cards: 136]

C199–C204: Doping, Diffusion & Photolithography

C199. Ion implantation: As^+ ions are accelerated through 50 keV and implanted into Si . The dose is $1.0 \times 10^{15}\text{ ions/cm}^2$ over a 200 mm wafer. How many grams of As ($M = 74.9$) are implanted per wafer? ($N_A = 6.022 \times 10^{23}$) [Cards: 4, 5]

C200. After implantation, the dopant is activated by rapid thermal annealing at 1000°C for 30 s. The diffusion coefficient of boron in Si at 1000°C is $D = 2.0 \times 10^{-14} \text{ cm}^2/\text{s}$. Using the diffusion length formula $L \approx \sqrt{Dt}$, estimate how far (in nm) boron atoms diffuse during the anneal. (1 nm = 10^{-7} cm) [Cards: 30]

C201. The diffusion of phosphorus in Si follows an Arrhenius temperature dependence: $D = D_0 e^{-E_a/(RT)}$ with $D_0 = 3.85 \text{ cm}^2/\text{s}$ and $E_a = 3.66 \text{ eV/atom}$. (1 eV = 96.5 kJ/mol. Use $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, and convert E_a to J/mol.) Calculate D at 1000°C . [Cards: 84, 85, 205]

C202. Photolithography uses a photoresist that reacts according to first-order kinetics under UV exposure. The exposure dose D (mJ/cm²) is related to the fraction f of unreacted resist by $f = e^{-kD}$, where $k = 0.030 \text{ cm}^2/\text{mJ}$. What dose is required for 99% of the resist to react ($f = 0.01$)? [Cards: 81]

C203. The photoresist from C202 has a contrast $\gamma = 3.0$, defined by $\gamma = \frac{1}{\log_{10}(D_0/D_1)}$ where D_0 is the dose to just clear the resist and D_1 is the dose where the resist just begins to clear. If $D_1 = 50 \text{ mJ/cm}^2$, what is D_0 ? [Cards: — (applied logarithm)]

C204. Developer solution for photoresist is typically 2.38% tetramethylammonium hydroxide (TMAH, $\text{N}(\text{CH}_3)_4\text{OH}$, $M = 91.2$) by mass. A fab uses 500 L of developer per day. The concentrated TMAH solution is 25% by mass (density 1.00 g/mL). What volume of concentrated TMAH is needed to prepare one day's supply of developer? (Assume developer density $\approx 1.00 \text{ g/mL}$.) [Cards: 133, 135, 137]

C205–C210: Advanced Device Fabrication & Materials

C205. Copper electroplating for damascene interconnects uses $\text{CuSO}_4/\text{H}_2\text{SO}_4$ bath. The half-reaction is $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$. A 200 mm wafer has 10^4 trenches, each $0.10 \text{ }\mu\text{m}$ wide, $0.30 \text{ }\mu\text{m}$ deep, and 1.0 cm long. What total charge (in coulombs) is required to fill all trenches with Cu (density 8.96 g/cm^3 , $M_{\text{Cu}} = 63.5$)? ($F = 96,485$) [Cards: 1, 129, 131, 132, 158]

C206. Chemical Mechanical Polishing (CMP) removes excess Cu after electroplating. The CMP slurry contains 5.0% H_2O_2 by mass (density 1.02 g/mL , $M_{\text{H}_2\text{O}_2} = 34.0$) to oxidize the Cu surface: $\text{Cu} + \text{H}_2\text{O}_2 \rightarrow \text{CuO} + \text{H}_2\text{O}$. If the slurry flow rate is 200 mL/min , how many moles of Cu can be oxidized per minute? (Assume H_2O_2 is the limiting reagent at the wafer surface.) [Cards: 1, 8, 9, 11, 133, 135]

C207. Hafnium oxide (HfO_2) is a high- κ gate dielectric deposited by ALD: $\text{HfCl}_4(\text{g}) + 2\text{H}_2\text{O}(\text{g}) \rightarrow \text{HfO}_2(\text{s}) + 4\text{HCl}(\text{g})$. The deposited film is 2.0 nm thick on a 300 mm wafer. Density of HfO_2 is 9.68 g/cm^3 , $M_{\text{HfO}_2} = 210.5$. (a) What mass of HfO_2 is on the wafer? (b) What mass of HfCl_4 ($M = 320.3$) was consumed? [Cards: 1, 8, 9, 10, 158]

C208. The gate dielectric in C207 requires an equivalent oxide thickness (EOT) of 0.50 nm . EOT relates the physical thickness $t_{\text{high-}\kappa}$ to the equivalent SiO_2 thickness: $\text{EOT} = t_{\text{high-}\kappa} \times \frac{\kappa_{\text{SiO}_2}}{\kappa_{\text{high-}\kappa}}$, where $\kappa_{\text{SiO}_2} = 3.9$ and $\kappa_{\text{HfO}_2} = 25$. What physical thickness of HfO_2 is needed to achieve this EOT? Does the 2.0 nm film from C207 meet the requirement? [Cards: — (applied)]

C209. Titanium nitride (TiN) is deposited as a diffusion barrier by reactive sputtering: $\text{Ti}(\text{s}) + \frac{1}{2}\text{N}_2(\text{g}) \rightarrow \text{TiN}(\text{s})$, $\Delta H^\circ = -338 \text{ kJ/mol}$. A sputtering target contains 5.00 kg of Ti ($M = 47.9$). (a) How many moles of N_2 gas are consumed to fully react the Ti target? (b) What volume of N_2 at STP is required? ($M_{\text{N}_2} = 28.0$) [Cards: 1, 6, 7, 8, 9, 39, 40]

C210. A state-of-the-art logic chip contains $\sim 5.0 \times 10^{10}$ transistors on a 300 mm wafer. Each transistor's gate is a fin of Si ($0.020 \mu\text{m}$ wide $\times$ $0.050 \mu\text{m}$ tall $\times$ $0.10 \mu\text{m}$ long, density of Si = 2.33 g/cm^3). (a) What is the total mass of Si in all the transistor fins on one wafer? (b) This Si was deposited epitaxially from SiH_4 (C183). What mass of SiH_4 ($M = 32.1$) was required, assuming 100% incorporation? (c) The fab processes 10,000 wafers per month. What is the annual SiH_4 consumption in kg? [Cards: 1, 4, 5, 8, 9, 10, 158]

ULTRA-INDUSTRIAL — 10 Problems (Level 5, 7–10 domains)

These are not "problems" in the ordinary sense. They are miniature chemical engineering design exercises. Each one chains 7 or more domains from the vocabulary deck. The Trigger Trace for a single problem may fire 12 cards across stoichiometry, gases, thermochemistry, kinetics, equilibrium, electrochemistry, solutions, and bridges. Solve one per day. They are meant to be hard.

C211–C215: Integrated Process Design — Full Plant Optimization

C211. [7 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Solutions + Bridges] A Haber-Bosch reactor operates at 450°C and 250 atm. The feed is a 3 : 1 H_2 : N_2 mixture at 25°C . (a) The feed must be preheated to 450°C before entering the reactor. The average C_p of the 3 : 1 H_2 : N_2 mixture is $29.1 \text{ J mol}^{-1} \text{ K}^{-1}$. How much heat (kJ) is required to preheat 1000 mol of feed gas? (b) At 450°C , $K_p = 4.3 \times 10^{-5} \text{ atm}^{-2}$ for $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$. The reactor achieves 85% of the equilibrium-limited conversion. Using an ICE table with initial $P_{\text{N}_2} = 62.5 \text{ atm}$ and $P_{\text{H}_2} = 187.5 \text{ atm}$, find the equilibrium P_{NH_3} , then the actual P_{NH_3} at 85% of that. (c) The exothermic reaction ($\Delta H^\circ = -92.2 \text{ kJ/mol}$) heats the product stream. If 15 mol of NH_3 are produced, how much heat is released? (d) This heat is recovered to generate steam. Water enters the heat exchanger at 25°C and exits as steam at 200°C ($c_{p,\text{water}} = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$, $c_{p,\text{steam}} = 2.00 \text{ J g}^{-1} \text{ K}^{-1}$, $\Delta H_{\text{vap}} = 40.7 \text{ kJ/mol}$ at 100°C , $M_{\text{H}_2\text{O}} = 18.0$). How many kg of steam can be generated from the heat in (c)? (e) The ammonia is condensed out at -33°C ($\Delta H_{\text{vap},\text{NH}_3} = 23.3 \text{ kJ/mol}$, $M_{\text{NH}_3} = 17.0$). How much additional cooling (kJ) is required to condense the 15 mol of NH_3 ? (f) Unreacted N_2 and H_2 are recycled. If the single-pass conversion of N_2 is 12%, use first-order recycle mathematics to find how many passes achieve 95% overall conversion. [Cards: 1, 8, 9, 19, 21, 26, 27, 33, 34, 38, 39, 40, 41, 48, 81, 89, 93, 95, 99, 158, 160, 162, 197, 199]

C212. [8 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Electrochem + Solutions + Bridges] An integrated methanol plant: Step 1 — Steam reforming: $\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3\text{H}_2$ ($\Delta H^\circ = +206 \text{ kJ/mol}$) at 850°C , 25 atm. Step 2 — Water-gas shift: $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$ ($\Delta H^\circ = -41.2 \text{ kJ/mol}$) at 250°C . Step 3 — Methanol synthesis: $\text{CO} + 2\text{H}_2 \rightleftharpoons \text{CH}_3\text{OH}$ ($\Delta H^\circ = -90.7 \text{ kJ/mol}$) at 250°C , 75 atm. (a) In Step 1, 1000 mol CH_4 and 3000 mol H_2O are fed. The equilibrium constant at 850°C is $K_p = 1.2 \times 10^4 \text{ atm}^2$. If 90% of CH_4 reacts, how many moles of H_2 exit the reformer? (b) The heat for the endothermic reforming is supplied by burning part of the CH_4 feed: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, $\Delta H^\circ = -890 \text{ kJ/mol}$. How many moles of CH_4 must be burned to supply the heat for Step 1, assuming 70% thermal efficiency? (c) In Step 2, the shift reactor at 250°C has $K_p = 2.8 \times 10^2$. With the outlet from (a), calculate Q and determine the equilibrium composition. (d) Before Step 3, the gas is compressed from 25 to 75 atm. Using $P_1 V_1 = P_2 V_2$ (isothermal compression at 250°C), by what factor does the volume decrease? (e) In Step 3, the methanol reactor at 250°C and 75 atm has $K_p = 2.5 \times 10^{-2} \text{ atm}^{-2}$. If the inlet $\text{CO} : \text{H}_2$ ratio is 1 : 2 and the total inlet pressure of reactants is

60 atm, set up the ICE table. (f) The crude methanol is purified by distillation. The methanol-water mixture (30% water by mass) is distilled. How much heat is needed to vaporize 100 kg of the mixture? ($\Delta H_{\text{vap,CH}_3\text{OH}} = 35.2 \text{ kJ/mol}$, $M = 32.0$; $\Delta H_{\text{vap,H}_2\text{O}} = 40.7 \text{ kJ/mol}$, $M = 18.0$) [Cards: 1, 8, 9, 10, 11, 19, 21, 26, 27, 33, 38, 39, 40, 41, 42, 48, 89, 90, 91, 93, 95, 135, 160, 162]

C213. [7 domains: Stoich + Gases + Thermo + Electrochem + Kinetics + Equilibrium + Bridges] The chlor-alkali membrane cell produces Cl_2 , H_2 , and NaOH . Downstream, Cl_2 is reacted with H_2 to make HCl : $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$, $\Delta H^\circ = -184.6 \text{ kJ/mol}$. The HCl is then oxidized back to Cl_2 via the Deacon process: $4\text{HCl} + \text{O}_2 \rightleftharpoons 2\text{Cl}_2 + 2\text{H}_2\text{O}$, $\Delta H^\circ = -114.4 \text{ kJ/mol}$, catalyzed by CuCl_2 at 450°C . (a) A chlor-alkali cell runs at 200,000 A with 95% current efficiency. How many kg of Cl_2 ($M = 71.0$) are produced per day? ($F = 96,485$) The half-reaction: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$. (b) The Cl_2 from (a) is fully converted to HCl . What mass of H_2 ($M = 2.02$) from the same cell is consumed? (Half-reaction: $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ — the H_2 and Cl_2 are produced in a 1:1 mole ratio.) (c) The HCl is oxidized via the Deacon process. At 450°C , $K_p = 58 \text{ atm}^{-1}$ for $4\text{HCl} + \text{O}_2 \rightleftharpoons 2\text{Cl}_2 + 2\text{H}_2\text{O}$. The feed is stoichiometric HCl and O_2 at 1.50 atm total initial pressure. Set up the ICE table and find the equilibrium conversion of HCl . (d) The Deacon process is exothermic. The reactor operates adiabatically and the temperature rises. The rate constant k follows Arrhenius with $E_a = 85 \text{ kJ/mol}$. If the inlet is at 450°C and the outlet reaches 520°C , by what factor does k increase? ($R = 8.314$) (e) Calculate ΔG° for the Deacon process at 450°C from K_p and explain why a catalyst is essential despite favorable thermodynamics. [Cards: 1, 8, 9, 10, 39, 40, 41, 48, 49, 51, 84, 85, 86, 88, 89, 93, 95, 121, 122, 125, 126, 129, 131, 132, 164, 165, 166]

C214. [8 domains: Stoich + Gases + Thermo + Acid-Base + Equilibrium + Kinetics + Solutions + Bridges] Ostwald process for nitric acid: Step 1 — $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$ (Pt/Rh catalyst, 900°C , $\Delta H^\circ = -905 \text{ kJ/mol}$). Step 2 — $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ ($\Delta H^\circ = -114 \text{ kJ/mol}$, homogeneous, third-order). Step 3 — $3\text{NO}_2 + \text{H}_2\text{O} \rightarrow 2\text{HNO}_3 + \text{NO}$ (absorption tower). (a) Step 1: 500 kg of NH_3 ($M = 17.0$) is oxidized per hour at 95% conversion. What mass of NO ($M = 30.0$) is produced per hour? (b) The oxidation of NO to NO_2 (Step 2) is third-order: $\text{Rate} = k[\text{NO}]^2[\text{O}_2]$ with $k = 7.1 \times 10^3 \text{ M}^{-2} \text{ s}^{-1}$ at 298 K. The reaction is run at 50°C where k is much slower (it actually decreases with T for this unusual reaction — E_a is negative: -4.6 kJ/mol). At 50°C , $k = 4.2 \times 10^3 \text{ M}^{-2} \text{ s}^{-1}$. If $[\text{NO}] = 0.010 \text{ M}$ and $[\text{O}_2] = 0.0050 \text{ M}$, what is the initial rate at 50°C ? (c) The NO_2 absorption (Step 3) produces 60% HNO_3 by mass. A storage tank holds 10,000 kg of this acid. The density of 60% HNO_3 is 1.37 g/mL, $M_{\text{HNO}_3} = 63.0$. What is the molarity of HNO_3 ? (d) The tail gas contains unabsorbed NO . This is removed by reacting with NH_3 : $4\text{NO} + 4\text{NH}_3 + \text{O}_2 \rightarrow 4\text{N}_2 + 6\text{H}_2\text{O}$ (selective catalytic reduction). If the tail gas contains 500 ppm NO by volume at 1.00 atm and 200°C , what is the partial pressure of NO in atm? (1 ppm = 10^{-6} by volume.) What mass of NH_3 is needed per 1000 m^3 of tail gas (measured at 200°C , 1.00 atm)? ($R = 0.08206$) (e) The nitric acid is neutralized with NH_3 to make ammonium nitrate fertilizer: $\text{HNO}_3 + \text{NH}_3 \rightarrow \text{NH}_4\text{NO}_3$. If all the acid from (a) is neutralized, what mass of NH_4NO_3 ($M = 80.0$) is produced per hour? [Cards: 1, 8, 9, 10, 11, 13, 19, 20, 26, 33, 39, 40, 76, 77, 78, 85, 86, 101, 107, 133, 134, 135, 160]

C215. [7 domains: Stoich + Gases + Thermo + Equilibrium + Kinetics + Solutions + Bridges] Integrated Gasification Combined Cycle (IGCC) with pre-combustion CO_2 capture: Coal ($\approx \text{C}$) is gasified: $\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$ ($\Delta H^\circ = +131 \text{ kJ/mol}$) at 1200°C , 30 atm. The syngas is shifted: $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$ ($\Delta H^\circ = -41.2 \text{ kJ/mol}$). CO_2 is removed by Selexol (physical solvent, absorbs CO_2 at high pressure). The remaining H_2 is burned in a gas turbine. (a) A gasifier processes 1000 kg of carbon ($M = 12.0$) per hour. How many moles of syngas ($\text{CO} + \text{H}_2$) are produced per hour, assuming complete conversion? (b) The syngas enters the shift reactor at 400°C and 30 atm. $K_p = 11.7$ at

400°C. Initially, the syngas from (a) has $\text{CO} : \text{H}_2\text{O} : \text{H}_2 : \text{CO}_2 = 1 : 3 : 1 : 0$ mole ratio (steam is added before shifting). Set up the ICE table and find the equilibrium composition. (c) CO_2 is removed with 95% efficiency by Selexol absorption. The absorbed CO_2 is released when the pressure is dropped to 1 atm. What volume of CO_2 (at 25°C, 1 atm) is captured per hour? (d) The remaining H_2 -rich gas is burned: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, $\Delta H^\circ = -571.6 \text{ kJ/mol}$. Calculate the hourly heat output in GJ. (e) The combustion produces steam at 600°C and 100 atm that drives a turbine. The steam's C_p is $2.10 \text{ J g}^{-1} \text{ K}^{-1}$. If the steam cools to 100°C in the turbine, how much energy is extracted per kg of steam? If the turbine-generator is 40% efficient, what is the electrical output in MW? (1 MW = 10^6 J/s) (f) The plant must meet a CO_2 emission limit of 100 kg CO_2/MWh . With the 95% capture from (c), does it comply? Calculate kg CO_2 emitted per MWh of electricity from (e). [Cards: 1, 8, 9, 10, 19, 20, 26, 27, 33, 34, 38, 39, 40, 41, 48, 89, 90, 93, 95, 96, 99, 160, 162, 172, 197]

C216–C220: Advanced Industrial Synthesis & Environmental Control

C216. [7 domains: Stoich + Gases + Thermo + Equilibrium + Acid-Base + Solutions + Bridges] Flue gas desulfurization (FGD) by the limestone-gypsum wet scrubbing process: SO_2 is absorbed into a limestone slurry, oxidized, and crystallized as gypsum. $\text{SO}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{aq})$, Henry's Law constant $K_H = 1.2 \text{ M/atm}$ at 55°C. $\text{SO}_2(\text{aq}) + \text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{HSO}_3^-$, $K_{a1} = 1.7 \times 10^{-2}$. $\text{CaCO}_3(\text{s}) + 2\text{H}^+ \rightarrow \text{Ca}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$. $\text{HSO}_3^- + \frac{1}{2}\text{O}_2 \rightarrow \text{SO}_4^{2-} + \text{H}^+$. $\text{Ca}^{2+} + \text{SO}_4^{2-} + 2\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}(\text{s})$ (gypsum). (a) A power plant exhausts $10^6 \text{ m}^3/\text{hr}$ of flue gas at 55°C and 1.00 atm containing 1500 ppm SO_2 by volume. What is the partial pressure of SO_2 in atm? (1 ppm = 10^{-6}) (b) Using Henry's Law, what is the equilibrium concentration of dissolved SO_2 at the gas-liquid interface? (c) The dissolved SO_2 dissociates into HSO_3^- (first dissociation dominates). From K_{a1} , calculate the equilibrium $[\text{H}^+]$ and pH at the interface, assuming $[\text{SO}_2(\text{aq})] = 1.8 \times 10^{-3} \text{ M}$ from (b). (d) CaCO_3 is added to neutralize the acid. What is the minimum mass of CaCO_3 ($M = 100.1$) required per hour to neutralize the H^+ produced, assuming all SO_2 ultimately produces 2 H^+ per SO_2 (from the overall stoichiometry: $\text{SO}_2 + \frac{1}{2}\text{O}_2 + \text{CaCO}_3 + 2\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O} + \text{CO}_2$)? (e) The gypsum produced is $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ ($M = 172.2$). What is the daily gypsum production in metric tons? (f) The scrubbing liquor is recycled. Lime is sometimes added to regenerate the limestone: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2$, $\Delta H^\circ = -65.2 \text{ kJ/mol}$. How much heat is released per ton of CaO ($M = 56.1$) slaked? [Cards: 1, 8, 9, 10, 11, 19, 26, 33, 39, 40, 41, 89, 93, 95, 101, 104, 107, 109, 133, 160]

C217. [7 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Solutions + Bridges] Fischer-Tropsch (FT) synthesis converts syngas to liquid hydrocarbons: $\text{CO} + 2\text{H}_2 \rightarrow -\text{CH}_2- + \text{H}_2\text{O}$ (simplified), $\Delta H^\circ \approx -165 \text{ kJ/mol CO}$. A cobalt catalyst at 220°C and 25 atm gives C_{5+} selectivity of 85%. (a) A FT reactor receives 5000 mol/hr of syngas with $\text{H}_2/\text{CO} = 2.0$. At 60% single-pass CO conversion, how many moles of H_2O are produced per hour? (b) The FT reaction is highly exothermic. The reactor tubes are cooled by boiling water at 250°C ($\Delta H_{\text{vap}} = 40.7 \text{ kJ/mol}$ modified for pressure). If the heat released is 165 kJ per mole of CO converted, how many kg of steam are generated per hour? ($M_{\text{H}_2\text{O}} = 18.0$) (c) The FT wax (C_{20+}) is hydrocracked to diesel: $\text{C}_{20}\text{H}_{42} + \text{H}_2 \rightarrow \text{C}_{10}\text{H}_{22} + \text{C}_{10}\text{H}_{22}$ (simplified). If 100 kg of wax ($M \approx 282$) yields 85 kg of diesel ($M \approx 142$), what is the percent yield? (d) The aqueous phase from FT contains 2.0% methanol and 1.0% ethanol by mass (density 1.00 g/mL). What is the molarity of methanol ($M = 32.0$) in this stream? (e) Unconverted syngas is recycled. If the recycle ratio (recycle flow / fresh feed flow) is 3.0, and the single-pass CO conversion is 60%, what is the overall CO conversion? (Use: overall = single-pass / $[1 - (1 - \text{single-pass}) \times (\text{recycle fraction})]$ where recycle fraction = recycle ratio / (1 + recycle ratio).) (f) The tail gas ($\text{CO} + \text{H}_2 + \text{CH}_4 + \text{CO}_2$) is burned in a flare. The tail gas has an average heating value of 250 kJ/mol. If 100 mol/hr of tail gas is flared, what is the heat output in MJ/hr, and what mass of CO_2 ($M = 44.0$) is

emitted if the average carbon number of the tail gas is 0.3 mol C per mol gas? [Cards: 1, 8, 9, 10, 13, 19, 33, 38, 39, 40, 41, 81, 89, 93, 95, 133, 134, 158, 160]

C218. [8 domains: Stoich + Gases + Thermo + Electrochem + Kinetics + Equilibrium + Solutions + Bridges]

Hydrogen economy: green H_2 from PEM electrolysis $\rightarrow$ compressed storage $\rightarrow$ fuel cell vehicle. (a) A PEM electrolyzer operates at 1.80 V and 500,000 A. How many kg of H_2 ($M = 2.02$) are produced per hour? ($F = 96,485$, half-reaction: $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$) (b) The electrolyzer consumes $1.80 \text{ V} \times 500,000 \text{ A} = 900 \text{ kW}$ of electrical power. The higher heating value (HHV) of H_2 is 286 kJ/mol. What is the energy efficiency (HHV of H_2 produced / electrical energy input)? (c) The H_2 is compressed from 1 atm to 700 atm at 25°C for storage. Using $P_1V_1 = P_2V_2$, what volume does 1.00 kg of H_2 occupy at 700 atm? (At 1 atm, 25°C , $n = 1000/2.02 = 495 \text{ mol}$, $V = nRT/P$.) Is the ideal gas assumption reasonable at 700 atm? Name the card that corrects for this (van der Waals). (d) The fuel cell: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, $E^\circ = 1.23 \text{ V}$. At an operating current of 200 A and voltage of 0.70 V, what is the fuel cell's voltage efficiency? ($\eta_V = V_{\text{operating}}/E^\circ$) (e) At 200 A, how many grams of H_2 are consumed per hour? (f) The fuel cell operates at 80°C . The Nernst equation predicts the thermodynamic potential at this temperature and at $P_{\text{H}_2} = P_{\text{O}_2} = 2.0 \text{ atm}$ (product water is liquid). Calculate the Nernst-adjusted potential: $E = E^\circ - \frac{RT}{nF} \ln \frac{1}{P_{\text{H}_2} P_{\text{O}_2}^{1/2}}$. ($R = 8.314$, $T = 353 \text{ K}$, $n = 2$) How does this compare to the actual operating voltage of 0.70 V? What causes the difference? [Cards: 1, 19, 20, 32, 39, 40, 48, 121, 122, 125, 126, 127, 128, 129, 131, 132, 160, 197]

C219. [7 domains: Stoich + Gases + Thermo + Equilibrium + Kinetics + Solutions + Bridges] Urea

production: $2\text{NH}_3 + \text{CO}_2 \rightleftharpoons \text{NH}_2\text{CONH}_2 + \text{H}_2\text{O}$ ($\Delta H^\circ = -117 \text{ kJ/mol}$). The reaction is run at 185°C and 180 atm with excess NH_3 . (a) A urea plant receives 500 kg/hr of NH_3 ($M = 17.0$) and 650 kg/hr of CO_2 ($M = 44.0$). Which is the limiting reagent? What is the theoretical urea production rate in kg/hr? ($M_{\text{urea}} = 60.1$) (b) The equilibrium constant at 185°C is $K = 1.5$ (based on molar concentrations in the liquid phase). The reactor achieves 75% of equilibrium conversion of the limiting reagent. What is the actual urea production rate? (c) Unreacted NH_3 and CO_2 are stripped and recycled. The stripper uses heat: incoming solution at 170°C must be heated to 200°C . The solution's C_p is $3.5 \text{ J g}^{-1} \text{ K}^{-1}$ and the flow rate is 2000 kg/hr. Heat is supplied by steam at 220°C ($\Delta H_{\text{vap}} = 38.0 \text{ kJ/mol}$ at pressure). How many kg of steam are needed per hour? ($M_{\text{H}_2\text{O}} = 18.0$) (d) The urea melt is prilled (sprayed into droplets that solidify). The melt enters at 140°C and cools to 80°C . The specific heat of urea melt is $2.0 \text{ J g}^{-1} \text{ K}^{-1}$, and the latent heat of solidification is 14.5 kJ/mol ($M_{\text{urea}} = 60.1$). For 1000 kg/hr of urea, what is the total heat that must be removed? (e) The prilling tower uses air at 25°C . The air exits at 60°C . Air's C_p is $1.01 \text{ J g}^{-1} \text{ K}^{-1}$. What mass of cooling air is needed per hour? ($\rho_{\text{air}} \approx 1.2 \text{ kg/m}^3$ at 25°C . What volume of air is this?) [Cards: 1, 8, 9, 10, 11, 12, 13, 14, 19, 33, 34, 38, 39, 40, 89, 93, 95, 135, 160]

C220. [7 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Electrochem + Solutions] Titanium dioxide (TiO_2) pigment via the Chloride Process: Step 1 — Carbochlorination:

$\text{TiO}_2(\text{s}) + 2\text{Cl}_2(\text{g}) + \text{C}(\text{s}) \rightarrow \text{TiCl}_4(\text{g}) + \text{CO}_2(\text{g})$ at 900°C , $\Delta H^\circ = -233 \text{ kJ/mol}$. Step 2 — Purification: TiCl_4 is distilled (bp 136°C , $\Delta H_{\text{vap}} = 36.0 \text{ kJ/mol}$, $M = 189.7$). Step 3 — Oxidation: $\text{TiCl}_4(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{TiO}_2(\text{s}) + 2\text{Cl}_2(\text{g})$ at 1400°C , $\Delta H^\circ = -179 \text{ kJ/mol}$. The Cl_2 is recycled. (a) Step 1: A reactor processes 500 kg/hr of TiO_2 ($M = 79.9$) with stoichiometric Cl_2 and excess C. What mass of Cl_2 ($M = 71.0$) is consumed per hour? (b) What mass of TiCl_4 ($M = 189.7$) is produced per hour at 95% yield? (c) The TiCl_4 is purified by distillation. How much heat is needed per hour to vaporize the TiCl_4 ? (d) Step 3: The TiCl_4 vapor is burned with O_2 in a flame reactor. The reaction produces TiO_2 particles and regenerates Cl_2 . What mass of Cl_2 is regenerated per hour? (The Cl_2 is recycled to Step 1 — what fraction of the Step 1 Cl_2 demand does this meet?) (e) The flame reactor operates at 1400°C .

Preheating the O_2 from 25°C to 1400°C requires heat. $C_{p,\text{O}_2} = 1.05 \text{ J g}^{-1} \text{ K}^{-1}$ (average over the range). For the O_2 needed in (d), what mass of O_2 is required, and how much preheat energy (kJ/hr) is needed? (f) The exothermic oxidation (Step 3) releases $179 \text{ kJ/mol TiCl}_4$. This heat could theoretically preheat the O_2 . Is the reaction exothermic enough to supply the preheat from (e)? Calculate the heat released per hour and compare. (g) The TiO_2 particles are collected. The product specification requires 99.0% TiO_2 purity. If the crude product contains 97.5% TiO_2 , how much impurity (kg per ton of crude) must be removed? [Cards: 1, 8, 9, 10, 11, 12, 13, 33, 38, 39, 40, 41, 42, 121, 122, 123, 160, 162, 197]

ULTRA-SEMICONDUCTOR — 10 Problems (Level 5, 7–10 domains)

Semiconductor manufacturing is the most chemically intensive industry per gram of product. A single chip passes through 300+ chemical steps. These problems chain gas delivery, surface kinetics, plasma physics, thermal budgets, and waste treatment into single multi-domain exercises.

C221–C225: Integrated Front-End Processes

C221. [7 domains: Stoich + Gases + Kinetics + Thermo + Equilibrium + Solutions + Bridges] Thermal oxidation of Si with HCl gettering and dopant redistribution. A batch furnace oxidizes 100 wafers (200 mm diameter) at 1000°C in dry O_2 with 3% HCl by volume. (a) The furnace tube volume is 50 L. At the process pressure of 1.00 atm and 1000°C , how many moles of gas are in the tube? ($R = 0.08206$) What are the partial pressures of O_2 and HCl? (b) The target oxide thickness is 50 nm (density 2.20 g/cm^3 , $M_{\text{SiO}_2} = 60.1$). What total mass of SiO_2 grows across all wafers? What mass of Si is consumed? ($M_{\text{Si}} = 28.1$) (c) The oxidation follows the linear-parabolic Deal-Grove model. In the linear regime, the rate is $dx/dt = k_L C_{\text{O}_2}$, where C_{O_2} is the O_2 concentration at the Si surface. C_{O_2} is governed by Henry's Law: $C = K_H P_{\text{O}_2}$ where $K_H = 5.0 \times 10^{-4} \text{ M/atm}$ at 1000°C . If the linear rate constant $k_L = 2.0 \times 10^{-4} \text{ cm}^2 \text{ mol}^{-1} \text{ s}^{-1}$, what is the initial oxidation rate in nm/min? ($1 \text{ M} = 10^{-3} \text{ mol/cm}^3$) (d) HCl gettering removes metal impurities by forming volatile metal chlorides. If the incoming O_2 contains 10 ppb Fe (by atoms relative to O_2), and all Fe is converted to FeCl_3 (bp 316°C), what mass of Fe ($M = 55.8$) enters the furnace per batch? (e) The HCl also affects the oxidation rate by forming H_2O : $4\text{HCl} + \text{O}_2 \rightleftharpoons 2\text{H}_2\text{O} + 2\text{Cl}_2$. At 1000°C , $K_p = 0.35$. If the initial $P_{\text{HCl}} = 0.03 \text{ atm}$ and $P_{\text{O}_2} = 0.97 \text{ atm}$, what is the equilibrium $P_{\text{H}_2\text{O}}$? H_2O accelerates oxidation — does this equilibrium produce significant H_2O ? [Cards: 1, 8, 9, 10, 19, 26, 27, 28, 76, 78, 80, 89, 93, 95, 158, 160, 186]

C222. [8 domains: Stoich + Gases + Kinetics + Thermo + Equilibrium + Electrochem + Solutions + Bridges] Plasma-enhanced CVD (PECVD) of Si_3N_4 from SiH_4 , NH_3 , and N_2 in a parallel-plate reactor at 300°C and 2.0 torr. The plasma dissociates the precursor gases into radicals, dramatically enhancing the deposition rate. (a) The overall reaction is $3\text{SiH}_4 + 4\text{NH}_3 \rightarrow \text{Si}_3\text{N}_4 + 12\text{H}_2$. The deposition rate is 200 nm/min on a 300 mm wafer (density of $\text{Si}_3\text{N}_4 = 3.44 \text{ g/cm}^3$, $M = 140.3$). What mass of Si_3N_4 is deposited per wafer per minute? (b) The precursor gases are fed at $\text{SiH}_4 : 50 \text{ sccm}$, $\text{NH}_3 : 200 \text{ sccm}$, $\text{N}_2 : 500 \text{ sccm}$ ($1 \text{ sccm} = 1 \text{ cm}^3/\text{min}$ at STP). What is the partial pressure of SiH_4 in the reactor (total $P = 2.0 \text{ torr}$)? (c) Only 5% of the SiH_4 fed is actually incorporated into the film (the rest is pumped away). What is the SiH_4 utilization efficiency? Verify: from (a), how many mol/min of Si are deposited? From (b), how many mol/min of SiH_4 are fed? (d) The plasma power is 500 W (J/s). If 40% of this power goes into dissociating NH_3 ($\text{N} - \text{H}$ bond energy: 391 kJ/mol), how many mol/min of NH_3 can be dissociated? Compare to the NH_3

feed rate from (b) — is the plasma power sufficient? (e) The effluent gas contains unreacted SiH_4 , which is pyrophoric (ignites in air). It is treated by a burn box: $\text{SiH}_4 + 2\text{O}_2 \rightarrow \text{SiO}_2 + 2\text{H}_2\text{O}$. If 95% of the SiH_4 is unreacted, what mass of SiO_2 ($M = 60.1$) accumulates in the burn box per day? (f) The burn box operates at 800°C . The heat from the SiH_4 combustion ($\Delta H^\circ_{\text{comb, SiH}_4} = -1517 \text{ kJ/mol}$) is recovered. How much heat (kJ/day) is recovered from the unreacted SiH_4 ? (g) The SiO_2 dust from the burn box must be disposed of. If the disposal cost is \$500 per metric ton, what is the annual disposal cost? [Cards: 1, 6, 7, 8, 9, 10, 11, 13, 19, 20, 26, 27, 33, 39, 40, 41, 42, 76, 78, 158, 160, 186]

C223. [7 domains: Stoich + Gases + Thermo + Equilibrium + Kinetics + Solutions + Bridges] Low-pressure CVD (LPCVD) of polysilicon from SiH_4 in a horizontal tube furnace at 620°C and 0.30 torr . The deposition is rate-limited by the surface reaction (first-order in SiH_4 partial pressure at the wafer surface). Gas-phase diffusion through the boundary layer also matters. (a) The furnace tube is 1.5 m long and 0.20 m in diameter. 100 wafers (200 mm diameter) are spaced 10 mm apart. What is the free volume in the tube (tube volume minus wafer volume)? (b) SiH_4 is fed at 100 sccm (STP). The total pressure is 0.30 torr ($= 4.0 \times 10^{-4} \text{ atm}$). Using $PV = nRT$, what is the volumetric flow rate at process conditions (620°C , 0.30 torr)? (c) The residence time $\tau = V/Q$ where V is the free volume and Q is the volumetric flow rate at process conditions. What is the residence time in seconds? (d) SiH_4 decomposes by first-order kinetics: $\text{Rate} = k[\text{SiH}_4]$ with $k = 0.15 \text{ s}^{-1}$ at 620°C (see C183). What fraction of SiH_4 decomposes during one pass through the furnace? ($f = 1 - e^{-k\tau}$) (e) The deposition rate is proportional to the SiH_4 partial pressure. The SiH_4 is depleted along the tube. If the inlet partial pressure is P_0 , the partial pressure at position x is $P(x) = P_0 e^{-kx/v}$ where v is the linear gas velocity. The velocity $v = Q/A_{\text{cross}}$, where A_{cross} is the open cross-sectional area of the tube (tube area minus wafer edge area). Calculate v (cm/s). Then find P/P_0 at $x = 75 \text{ cm}$ (halfway). (f) To compensate for depletion, the furnace temperature is ramped $+10^\circ\text{C}$ from inlet to outlet. Using Arrhenius with $E_a = 160 \text{ kJ/mol}$ (for polysilicon deposition from SiH_4), by what factor does k increase over this 10°C ramp? ($R = 8.314$) Does this compensate for the depletion in (e)? (g) The deposited film thickness uniformity must be within $\pm 3\%$ across all wafers. Based on the depletion profile from (e), is a temperature ramp necessary, or can the process work without it? [Cards: 1, 6, 7, 8, 19, 20, 21, 23, 26, 30, 76, 81, 83, 84, 85, 158, 160, 186, 197]

C224. [7 domains: Stoich + Gases + Thermo + Kinetics + Acid-Base + Equilibrium + Solutions] Post-etch residue removal uses a sequence of wet cleans. After a plasma etch that leaves fluorocarbon polymer residue ($\approx \text{CF}_2$), wafers undergo: (1) O_2 plasma ash to remove polymer; (2) dilute HF dip to remove damaged oxide; (3) SC1 ($\text{NH}_4\text{OH}/\text{H}_2\text{O}_2/\text{H}_2\text{O}$) to remove particles; (4) SC2 ($\text{HCl}/\text{H}_2\text{O}_2/\text{H}_2\text{O}$) to remove metals. (a) The O_2 plasma ash converts the polymer to CO_2 and SiF_4 (from underlying Si etch). If the polymer is $(\text{CF}_2)_n$ with $n \approx 1000$, and the total polymer mass per wafer is 0.50 mg , write the idealized ash reaction: $(\text{CF}_2)_n + \text{O}_2 \rightarrow \text{CO}_2 + 2\text{F}$ (F atoms then form SiF_4 with Si from the wafer). What volume of O_2 at STP is consumed per wafer? (b) The HF dip is $100 : 1 \text{ H}_2\text{O} : \text{HF}$ (by volume) using 49% HF stock. The final solution is 0.50% HF by mass. What is the molarity of HF in the dip? (Density 1.00 g/mL , $M_{\text{HF}} = 20.0$) The dip removes 2.0 nm of thermally grown SiO_2 (density 2.20 g/cm^3) from a 300 mm wafer. How many moles of HF are consumed? ($\text{SiO}_2 + 6\text{HF} \rightarrow \text{H}_2\text{SiF}_6 + 2\text{H}_2\text{O}$) (c) The SC1 solution ($1 : 1 : 5 \text{ NH}_4\text{OH}/\text{H}_2\text{O}_2/\text{H}_2\text{O}$) is prepared at 75°C . At this temperature, H_2O_2 decomposes: $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, first-order with $t_{1/2} = 8.0 \text{ min}$ at 75°C . If the initial $[\text{H}_2\text{O}_2] = 1.5 \text{ M}$, what is the concentration after 30 min ? How often should the bath be recharged? (d) The SC2 solution ($1 : 1 : 6 \text{ HCl}/\text{H}_2\text{O}_2/\text{H}_2\text{O}$) is at 75°C . The high temperature and acidity cause accelerated H_2O_2 decomposition. The decomposition is catalyzed by H^+ : $\text{Rate} = k[\text{H}_2\text{O}_2][\text{H}^+]$ with $k = 1.8 \times 10^{-4} \text{ M}^{-1} \text{ s}^{-1}$ at 75°C . If $[\text{HCl}] = 1.5 \text{ M}$ and initial $[\text{H}_2\text{O}_2] = 1.2 \text{ M}$, what is the half-life of H_2O_2 in SC2? Compare to SC1 — why is SC2 much less stable? (e) The waste from all these cleaning steps is combined and neutralized with

$\text{Ca}(\text{OH})_2$ before discharge. The combined waste contains 0.50 mol HF, 0.20 mol HCl, and 0.10 mol NH_4OH per wafer batch (25 wafers). What mass of $\text{Ca}(\text{OH})_2$ ($M = 74.1$) is needed per batch? (Neutralization: $\text{Ca}(\text{OH})_2 + 2\text{H}^+ \rightarrow \text{Ca}^{2+} + 2\text{H}_2\text{O}$; calculate total moles of H^+) [Cards: 1, 8, 9, 10, 11, 19, 33, 76, 81, 83, 86, 101, 107, 108, 109, 133, 134, 158, 160]

C225. [8 domains: Stoich + Gases + Thermo + Equilibrium + Kinetics + Electrochem + Solutions + Bridges]

Ion implantation of arsenic into silicon, followed by rapid thermal annealing (RTA) for dopant activation and damage repair, then electrochemical measurement of sheet resistance. (a) As^+ ions are implanted at 50 keV with a dose of 5.0×10^{15} ions/ cm^2 into a 300 mm Si wafer. What is the total number of As atoms implanted? What mass of As ($M = 74.9$) is this? (b) The implanted As atoms are initially mostly interstitial (not electrically active). RTA at 1050°C for 10 s activates 98% of the dopant. The diffusion coefficient of As in Si at 1050°C follows $D = D_0 e^{-E_a/(RT)}$ with $D_0 = 0.32 \text{ cm}^2/\text{s}$ and $E_a = 3.43 \text{ eV}$. ($1 \text{ eV} = 96.5 \text{ kJ/mol}$, $R = 8.314$) Calculate D and the diffusion length $\sqrt{Dt}$. Does the dopant profile broaden significantly during the 10 s anneal? (c) After activation, the sheet resistance R_s is measured by a four-point probe. $R_s = \rho/t$ where $\rho = 1/(qn\mu)$ is the resistivity. $q = 1.602 \times 10^{-19} \text{ C}$ (electron charge), n is the carrier concentration (activated As atoms per cm^3), and $\mu = 50 \text{ cm}^2/\text{V s}$ is the electron mobility. The implant depth is approximately 0.10 μm . Calculate the carrier concentration n (atoms/ cm^3), the resistivity ρ , and the sheet resistance R_s in $\Omega/\square$. (d) A separate test structure is a Van der Pauw cross for Hall effect measurement. Applying a current of 1.00 mA through the sample from (c) and a perpendicular magnetic field of 0.50 T produces a Hall voltage V_H . The Hall coefficient $R_H = V_H t / (IB) = 1/(nq)$. Calculate R_H and V_H . (e) The RTA chamber is purged with N_2 gas at 1.00 atm. Before annealing, the chamber (volume 5.0 L) must have O_2 reduced below 1.0 ppm. Initially, the chamber is full of air (21% O_2). How many purge cycles (evacuate to 0.01 atm and refill with N_2 to 1.00 atm) are needed to reach < 1.0 ppm O_2 ? (Each cycle reduces O_2 by the pressure ratio: $P_{\text{O}_2, \text{new}} = P_{\text{O}_2, \text{old}} \times (0.01/1.00)$.) (f) After RTA, the wafer is rapidly cooled from 1050°C to 100°C in 20 s. The wafer's heat capacity is approximately that of Si: $C_p = 0.705 \text{ J g}^{-1} \text{ K}^{-1}$, wafer mass = 128 g (300 mm, 0.775 mm thick, $\rho_{\text{Si}} = 2.33 \text{ g/cm}^3$). Verify the wafer mass. What is the average cooling rate in K/s? How much heat (kJ) must be removed? [Cards: 1, 4, 5, 19, 21, 26, 33, 81, 84, 85, 158, 186, 197, 204, 205]

C226–C230: Integrated Back-End & Advanced Patterning

C226. [7 domains: Stoich + Gases + Thermo + Kinetics + Electrochem + Solutions + Bridges] Copper dual-damascene: (1) Barrier/seed PVD of TaN/Ta and Cu seed; (2) Cu electroplating to fill vias and trenches; (3) CMP to remove excess Cu; (4) Post-CMP clean. A 300 mm wafer has 10^{10} vias (each 0.050 μm diameter, 0.50 μm deep) and 5×10^9 trenches (0.050 μm wide, 0.20 μm deep, average length 5.0 μm). (a) Calculate the total volume of Cu to fill all features on one wafer. (b) Cu electroplating: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$. The plating current is 10.0 A for a 300 mm wafer. Theoretically (at 100% current efficiency), how long (in seconds) to deposit the Cu volume from (a)? ($\rho_{\text{Cu}} = 8.96 \text{ g/cm}^3$, $M_{\text{Cu}} = 63.5$, $F = 96,485$) (c) In practice, the plating also deposits a 0.50 μm "overburden" of Cu across the entire wafer surface. What additional Cu volume and plating time does this require? (d) The CMP step removes the overburden and the barrier layer. The CMP slurry contains 5.0% H_2O_2 (see C206) and abrasive SiO_2 nanoparticles (50 nm diameter, density 2.20 g/cm^3). The slurry has 1.0×10^{14} particles per mL. What is the mass percent of SiO_2 in the slurry? (e) After CMP, the wafer is cleaned with a solution of 0.50% HF + 1.0% citric acid ($\text{C}_6\text{H}_8\text{O}_7$, a weak triprotic acid, $\text{p}K_{a1} = 3.13$) by mass. What is the pH of the citric acid component? (Assume only the first dissociation matters, $M_{\text{citric}} = 192.1$, density 1.00 g/mL .) (f) The CMP waste slurry contains 100 mg/L Cu. Before discharge, Cu must be below 1.0 mg/L. The waste (1000 L/day) is treated by electroplating the Cu onto a sacrificial cathode at 50 A. How many hours per day must the

treatment cell operate? ($F = 96,485$) [Cards: 1, 4, 5, 8, 9, 10, 19, 76, 101, 104, 109, 121, 122, 129, 131, 132, 133, 134, 135, 141, 142, 158, 160]

C227. [7 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Solutions + Bridges] Atomic Layer Deposition (ALD) of HfO_2 high- κ dielectric using HfCl_4 and H_2O precursors in a cross-flow reactor at 300°C and 0.50 torr. ALD is a surface-saturated, self-limiting process: each precursor pulse deposits exactly one sub-monolayer. (a) The ALD half-reactions: Pulse A: $\text{Si-OH}_{(\text{s})} + \text{HfCl}_{4(\text{g})} \rightarrow \text{Si-O-HfCl}_{3(\text{s})} + \text{HCl}_{(\text{g})}$. Pulse B: $\text{HfCl}_{3(\text{s})} + \text{H}_2\text{O}_{(\text{g})} \rightarrow \text{Hf-OH}_{(\text{s})} + \text{HCl}_{(\text{g})}$. The net reaction per cycle is $\text{HfCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{HfO}_2 + 4\text{HCl}$. Each cycle deposits 0.12 nm of HfO_2 (density 9.68 g/cm³, $M = 210.5$). For a 300 mm wafer, how many Hf atoms are deposited per cm² per cycle? ($N_A = 6.022 \times 10^{23}$) (b) A typical ALD cycle: HfCl_4 pulse 0.50 s, N_2 purge 5.0 s, H_2O pulse 0.30 s, N_2 purge 5.0 s. What is the cycle time? To deposit a 3.0 nm film, how many cycles and what total process time? (c) The HfCl_4 precursor is a solid at room temperature ($P_{\text{vap}} = 0.10$ torr at 150°C). The precursor bottle is heated to 150°C . The vapor is transported to the reactor by 50 sccm of N_2 carrier gas. Using Dalton's Law, what is the molar flow rate of HfCl_4 ? (Total P at the bottle = 1.0 atm; the carrier gas saturates with HfCl_4 vapor. $P_{\text{HfCl}_4}/P_{\text{total}} = n_{\text{HfCl}_4}/n_{\text{total}}$.) (d) The HfCl_4 utilization (fraction of precursor that reacts on the wafer) is only 2.0% . How much HfCl_4 ($M = 320.3$) is consumed per wafer (wasted + used) for the 3.0 nm film? (e) The unreacted HfCl_4 and HCl byproduct are trapped in a scrubber containing CaCO_3 : $2\text{HCl} + \text{CaCO}_3 \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$. How many grams of CaCO_3 ($M = 100.1$) are consumed per 100 wafers? (HfCl_4 hydrolyzes in the scrubber: $\text{HfCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{HfO}_2 + 4\text{HCl}$, so total HCl = from precursor + from reaction.) (f) To improve throughput, the N_2 purge time is reduced from 5.0 to 2.0 s. However, if CVD-like reactions occur in the gas phase (from inadequate purging), particles form. The particle formation rate follows Arrhenius: $E_a = 85$ kJ/mol, and is first-order in residual precursor concentration. If purging follows first-order decay with time constant $\tau = 1.0$ s, the residual concentration after 5.0 s is $C_0 e^{-5}$. After 2.0 s, it is $C_0 e^{-2}$. By what factor does the particle formation rate increase when reducing purge from 5.0 to 2.0 s? [Cards: 1, 8, 9, 10, 19, 26, 27, 76, 81, 83, 84, 85, 133, 158, 160, 186]

C228. [8 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Solutions + Acid-Base + Bridges] Extreme ultraviolet (EUV) lithography uses 13.5 nm photons generated by a laser-produced plasma from Sn droplets. (a) The EUV source vaporizes Sn ($M = 118.7$) droplets. A droplet of 30 μm diameter ($\rho_{\text{Sn}} = 7.29$ g/cm³) is hit by a laser. How many Sn atoms are in one droplet? (b) The Sn plasma produces EUV light by $10\times$ ionizing Sn. The ionization energy to reach Sn^{10+} from neutral Sn is approximately the sum of the first 10 ionization energies: $\text{IE}_1 = 709$, $\text{IE}_2 = 1412$, $\text{IE}_3 = 2943$, $\text{IE}_4 = 3930$, $\text{IE}_5 = 7460$, $\text{IE}_6 = 8600$, $\text{IE}_7 = 9900$, $\text{IE}_8 = 11200$, $\text{IE}_9 = 12500$, $\text{IE}_{10} = 13800$ kJ/mol. What is the total energy (kJ/mol) to produce Sn^{10+} ? What fraction of the laser pulse energy (200 mJ per droplet) does this represent? ($N_A = 6.022 \times 10^{23}$) (c) Sn debris coats the collector mirror. The mirror is cleaned by hydrogen radical etching: $\text{Sn}_{(\text{s})} + 4\text{H}\cdot_{(\text{g})} \rightarrow \text{SnH}_{4(\text{g})}$. $\text{H}\cdot$ is generated by flowing H_2 through a hot filament (2000°C). The equilibrium $\text{H}_2 \rightleftharpoons 2\text{H}$ has $K_p = 1.2 \times 10^{-3}$ at 2000°C . If $P_{\text{H}_2} = 0.10$ atm at the filament, what is P_{H} ? What fraction of H_2 is dissociated? (d) The EUV photoresist is a chemically amplified resist (CAR). Upon exposure, a photoacid generator (PAG) produces H^+ : $\text{PAG} + h\nu \rightarrow \text{H}^+ + \text{byproducts}$. The quantum yield is 0.30 (0.30 H^+ per photon). The EUV dose is 20 mJ/cm². The photon energy $E = hc/\lambda$ where $h = 6.626 \times 10^{-34}$ J s, $c = 3.00 \times 10^8$ m/s, $\lambda = 13.5 \times 10^{-9}$ m. (e) Calculate the number of photons per cm² in the dose. Then calculate $[\text{H}^+]$ generated in the resist (thickness 50 nm, density 1.0 g/cm³). The resist contains 5.0% PAG by mass ($M_{\text{PAG}} = 500$). (f) During post-exposure bake (PEB) at 110°C , the H^+ catalyzes a deprotection reaction that switches the resist solubility. The deprotection is first-order in $[\text{H}^+]$ with $k = 2.0 \times 10^3$ M⁻¹ s⁻¹ at 110°C . The PEB time is 60 s. What fraction of the protecting groups is removed? (Assume pseudo-first-order: $k' = k[\text{H}^+]$,

fraction = $1 - e^{-k't}$.) (g) The exposed resist is developed in 2.38% TMAH (see C204). After development, the pattern quality is assessed. The line-width roughness (LWR) is 3.0 nm (3σ). If the target critical dimension (CD) is 20 nm, what is the LWR as a percentage of CD? [Cards: 1, 4, 5, 19, 20, 26, 33, 39, 53, 60, 76, 81, 83, 89, 93, 95, 101, 104, 107, 109, 133, 135, 137, 158]

C229. [7 domains: Stoich + Gases + Thermo + Kinetics + Electrochem + Solutions + Bridges] Chemical vapor deposition of tungsten (W) for contact plugs using WF_6 and H_2 in a cold-wall reactor at 400°C , 30 torr. Tungsten CVD is used to fill high-aspect-ratio contacts because of its excellent conformality. (a) The overall reaction is $\text{WF}_6(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow \text{W}(\text{s}) + 6\text{HF}(\text{g})$, $\Delta H^\circ = +69 \text{ kJ/mol}$ (actually endothermic for this specific reduction; the Si reduction is exothermic). What does the positive ΔH° imply about temperature control? Why must the reactor be heated despite the reaction being endothermic? (b) A 300 mm wafer has 5.0×10^9 contacts (each 0.10 μm diameter, 1.0 μm deep). What total volume and mass of W ($\rho = 19.3 \text{ g/cm}^3$, $M = 183.8$) is needed? (c) The deposition rate is limited by WF_6 supply at low pressures. The feed gas is 10% WF_6 , 30% H_2 , 60% Ar at 30 torr total. What is P_{WF_6} ? (d) The sticking coefficient of WF_6 on W at 400°C is $\gamma = 0.10$ (probability that an impinging molecule reacts). The impingement rate (molecules per cm^2 per second) from kinetic theory is $Z = \frac{P}{\sqrt{2\pi MRT}}$, with P in Pa ($1 \text{ atm} = 101,325 \text{ Pa}$), M in kg/mol ($M_{\text{WF}_6} = 0.2978 \text{ kg/mol}$), $R = 8.314$, T in K. Calculate Z and the deposition rate in nm/s. ($\rho_{\text{W}} = 19.3 \text{ g/cm}^3$; convert molecules/ cm^2/s to $\text{g/cm}^2/\text{s}$ to nm/s.) (e) The byproduct HF etches SiO_2 : $\text{SiO}_2 + 6\text{HF} \rightarrow \text{H}_2\text{SiF}_6 + 2\text{H}_2\text{O}$. The reactor wall has a quartz (SiO_2) liner. If 5.0 g of W are deposited per wafer, how much HF is produced? What mass of the quartz liner is etched per wafer? ($M_{\text{SiO}_2} = 60.1$) (f) The effluent HF must be scrubbed. A wet scrubber uses NaOH: $\text{HF} + \text{NaOH} \rightarrow \text{NaF} + \text{H}_2\text{O}$. If the fab runs 1000 wafers per day through this process, what mass of NaOH ($M = 40.0$) is consumed daily? (g) Excess WF_6 is also scrubbed: $\text{WF}_6 + 8\text{NaOH} \rightarrow \text{Na}_2\text{WO}_4 + 6\text{NaF} + 4\text{H}_2\text{O}$. Only 15% of the WF_6 is consumed in the reactor (the rest is pumped out). What additional NaOH is needed per day? Total NaOH consumption? [Cards: 1, 8, 9, 10, 11, 19, 20, 26, 27, 30, 33, 39, 40, 41, 76, 78, 84, 101, 107, 108, 129, 131, 133, 158, 160, 197, 199]

C230. [10 domains: Stoich + Gases + Thermo + Kinetics + Equilibrium + Electrochem + Acid-Base + Solutions + Bridges + Unit Conversions] End-to-end chip manufacturing environmental footprint. A logic fab processes 50,000 wafers (300 mm) per month. Calculate the annual environmental metrics using the full vocabulary deck. (a) **[Stoich + Gases]** The fab's N_2 consumption (for purging) is 500 m^3 per wafer (measured at STP). What is the annual N_2 consumption in metric tons? ($M_{\text{N}_2} = 28.0$, $1 \text{ m}^3 = 1000 \text{ L}$) (b) **[Gases + Thermo]** The N_2 is generated on-site by cryogenic air separation. Air is cooled from 25°C to -196°C (77 K). Air is 78% N_2 by volume, $C_{p,\text{air}} = 1.01 \text{ J g}^{-1} \text{ K}^{-1}$, $M_{\text{air}} \approx 29.0$. What is the theoretical minimum energy (GJ/yr) to cool the air needed for the annual N_2 production? (Ignore latent heats of O_2 and Ar condensation for this estimate.) (c) **[Electrochem + Stoich]** The fab's ultrapure water (UPW) consumption is 2000 L per wafer. UPW is produced by reverse osmosis followed by electrodeionization (EDI). EDI uses 0.50 kWh/ m^3 of electricity. The electricity is generated from natural gas in a combined-cycle plant at 55% efficiency. Natural gas ($\approx \text{CH}_4$) combustion: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, $\Delta H^\circ = -890 \text{ kJ/mol}$. How many metric tons of CO_2 ($M = 44.0$) are emitted annually just from the EDI electricity? ($1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$) (d) **[Acid-Base + Solutions + Stoich]** The fab uses 100 L of 49% HF (density 1.18 g/mL) per 1000 wafers for etching and cleaning. All fluoride must be precipitated as CaF_2 ($K_{sp} = 3.9 \times 10^{-11}$) before discharge. The waste treatment adds $\text{Ca}(\text{OH})_2$: $2\text{HF} + \text{Ca}(\text{OH})_2 \rightarrow \text{CaF}_2 + 2\text{H}_2\text{O}$. What is the annual consumption of $\text{Ca}(\text{OH})_2$ ($M = 74.1$) in metric tons? The discharge limit is 15 mg/L fluoride. If the total wastewater flow is 5000 m^3 per day, does CaF_2 precipitation achieve the limit? (Calculate $[\text{F}^-]$ in equilibrium with CaF_2 , assuming excess Ca^{2+} at 0.010 M.) (e) **[Thermo + Gases + Kinetics]** The fab's

cleanroom HVAC circulates $10^7 \text{ m}^3/\text{hr}$ of air. The air is filtered and conditioned from outside conditions (35°C , 80% RH) to cleanroom conditions (22°C , 45% RH). The enthalpy change is approximately 40 kJ per kg of dry air. Air density at 22°C is $1.20 \text{ kg}/\text{m}^3$. What is the annual HVAC energy consumption in GJ? If this energy comes from the same natural gas plant as (c), what are the associated CO_2 emissions? (f)

[Equilibrium + Thermo + Solutions] The fab's CO_2 footprint from (a-e) totals to X metric tons/yr. A post-combustion amine capture plant is installed that captures 90% of the CO_2 from the natural gas flue. The capture solvent (MEA, monoethanolamine, $\text{HOCH}_2\text{CH}_2\text{NH}_2$) has a CO_2 loading capacity of 0.50 mol CO_2 per mol MEA. MEA's $M = 61.1$, density $1.01 \text{ g}/\text{mL}$. What volume of MEA circulates through the capture plant annually? (Assume MEA is regenerated and recycled, volume = one-time fill.) (g)

[Unit Conversions] Summarize: Express the fab's annual CO_2 emissions per wafer and per cm^2 of silicon produced (wafer area = $\pi \times (15 \text{ cm})^2$). [Cards: 1, 6, 7, 8, 9, 10, 11, 17, 19, 20, 26, 27, 33, 34, 38, 39, 40, 41, 42, 48, 51, 89, 93, 95, 101, 103, 104, 107, 108, 121, 122, 129, 131, 132, 133, 134, 135, 136, 137, 141, 142, 143, 144, 158, 160, 162, 186, 197, 198, 199, 201, 202, 203]

Next step: Test yourself. Pick one problem from each section. Solve using the method in [chemistry-without-intuition.md](#). Check your answers against [chemistry-answers.md](#). Repeat until every card reference is fluent.